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CYBER WARFARE OPERATIONS



CAREER FIELD EDUCATION AND TRAINING PLAN

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CAREER FIELD EDUCATION AND TRAINING PLAN

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PART I

Preface

1. This Career Field Education and Training Plan (CFETP) is a comprehensive education and training document that identifies life-cycle education/training requirements, training support resources and minimum core task requirements for this specialty. The CFETP will provide personnel a clear career path to success and instill rigor in all aspects of career field training.

2. The CFETP documents the career field training program and consists of two parts. Management uses both parts in conjunction with myLearning to plan, manage, and control training within the career field. **NOTE:** Civilians occupying associated positions will use Part II to support duty position qualification training.

2.1. Part I provides information necessary for overall management of the specialty. Section A explains how to implement; Section B identifies career field progression information, duties and responsibilities, training strategies, and the career field path; Section C associates each level with specialty qualifications (knowledge, education, experience, training and other); Section D indicates resource constraints (e.g., funds, manpower, equipment, facilities); and Section E identifies transition training guide requirements for SSgt through MSgt.

2.2. Part II includes the following: Section A identifies the Specialty Training Standard (STS) and includes duties, tasks, Training References (TRs) to support training, AETC-conducted training, wartime course and core task and correspondence course requirements. Section B contains the Course Objectives List (COL) and training standards supervisors will use to determine if Airmen satisfied training requirements. Section C identifies available support materials (e.g., Qualification Training Package, which may be developed to support proficiency training). Section D identifies a training course index supervisors can use to determine resources available to support training. Included here are both mandatory and optional courses; and Section E identifies MAJCOM-unique training requirements supervisors can use to determine additional training required for the associated qualification needs. At unit level, supervisors and trainers will use Part II to identify, plan and conduct training commensurate with the overall goals of this plan.

3. Use of the guidance provided in this CFETP provides the foundation for effective and efficient training for individuals in this career field at the appropriate points in their careers. This plan enables the Air Force to train today's work force for tomorrow's jobs.

Abbreviations/Terms Explained

This section provides a common understanding of the terms that apply to the Cyber Warfare Operations CFETP.

Advanced Training (AT). A formal course of training that leads to a technical or supervisory level of an Air Force Specialty (AFS). Training is for selected Airmen at the advanced level of an AFS.

Air Education Training Command (AETC). Responsible for the recruiting, training and education of Air Force personnel. AETC also provides pre-commissioning, professional military and continuing education.

Air Force Career Field Manager (AFCFM). Representative appointed by the respective HQ USAF Deputy Chief of Staff or Under Secretariat to ensure that assigned Air Force specialties are trained and utilized to support Air Force mission requirements.

Air Force Job Qualification Standard (AFJQS). A comprehensive task list that describes a particular job type or duty position. Supervisors use the AFJQS to document task qualification. The tasks on AFJQSs are common to all persons serving in the described duty position.

Air Force Qualification Training Package (AFQTP). An instructional course designed for use at the unit to qualify or aid qualification in a duty position, program, or on a piece of equipment. It may be printed, computer-based, or other audiovisual media.

Air Force Specialty (AFS). A group of positions (with the same title and code) that require common qualifications.

Air Force Tactics, Techniques and Procedures (AFTTP). AFTTPs describe the proper employment of specific Air Force assets, individually or in concert with other assets, to accomplish detailed objectives.

Air University Associate-to-Baccalaureate Cooperative (AU ABC). Allows Airmen to turn a Community College of the Air Force Associates Degree into a Bachelor's Degree from an accredited university. The ABC program has established a partnership with various civilian higher-education institutions to offer four-year degree opportunities via distance learning. The participating schools will accept all of the credits earned by Airmen who have attained a CCAF degree and apply them to a Bachelor's degree related to their Air Force specialty.

Air University. Provides a full spectrum of Air Force education, offering graduate programs, professional development for officers, enlisted personnel, and civilians, and specialized courses for senior leaders; provides access to the Extension Course Institute.

Career Field Education and Training Plan (CFETP). A CFETP is a comprehensive core training document that identifies: life-cycle education and training requirements; training support resources; and minimum core task requirements for a specialty. The CFETP aims to give personnel a clear path and instill a sense of industry in career field training. CFETPs are officially posted at https://www.e-publishing.af.mil/.

Certification. A formal indication of an individual's ability to perform a task to required standards.

Cyber Effects Operations (CEO). Delivering an effect within the cyberspace war fighting domain.

Chief Enlisted Manager (CEM). Chief Master Sergeants that have extensive experience and training, and demonstrated managerial ability to plan, direct, coordinate, implement, and control a wide range of work activity. Some managerial duties and responsibilities that are common to all chief enlisted managers are: managing and directing personnel resource activities; interpreting and enforcing policy and applicable directives; establishing control procedures to meet work goals and standards; recommending or initiating actions to improve functional operation efficiency; planning and programming work commitments and schedules; developing plans regarding facilities, supplies, and equipment procurement and maintenance.

Command Line Interface (CLI). A command-line interface (CLI) is a means of interaction with a computer program where the user (or client) issues commands to the program in the form of successive lines of text (command lines).

Computer Based Training (CBT). A forum for training in which the student learns via a computer terminal. It is an especially effective training tool that allows the students to practice applications while they learn.

Core Task. A task AFCFMs identify as a minimum qualification requirement for everyone within an AFSC regardless of duty position. Core tasks may be specified for a particular skill level or in general across the AFSC. Guidance for using core tasks can be found in the applicable CFETP narrative.

Course Objective List (COL). A publication derived from initial/advanced skills Course Training Standard (CTS), identifying the tasks and knowledge requirements and respective standards provided to achieve a 3-skill level in this career field. Supervisors use the COL to assist in conducting graduate evaluations.

Course Training Standard (CTS). A standard developed for all courses not governed by an STS, including specialized training packages and computer-based training courses.

Critical Tasks. Critical Tasks are tasks that require specific training and certification above and beyond other tasks. Tasks may be defined as critical either through AFI, Technical Orders, higher headquarters, or at any level in the unit.

Cross-Utilization Training. Training on non-duty AFSC specific tasks.

Defensive Cyberspace Operations (DCO). Passive and active cyberspace operations intended to preserve the ability to utilize friendly cyberspace capabilities and protect data, networks, net-centric capabilities, and other designated systems by defeating on-going or imminent malicious cyber activities/actors (DOD Dictionary of Military and Associated Terms).

Department of the Air Force Enlisted Classification Directory (DAFECD). The official directory for all military enlisted classification descriptions, codes, and identifiers. Establishes the occupational structure of the Air Force enlisted force. The occupational structure is flexible to permit enlisted personnel to specialize and develop their skills and abilities while allowing the Air Force to meet changing mission requirements. Individual enlisted personnel have a joint responsibility with commanders and supervisors at all levels to fully develop their abilities consistent with Air Force needs and within the established patterns of specialization.

Department of Defense Cyber Workforce Framework (DCWF) The DCWF is the Department's standardized cyber workforce lexicon of work roles that will be used for coding positions. Workforce coding is used to support a number of DoD-wide workforce management and workforce planning activities.

Direct Reporting Unit (DRU). Air Force subdivisions directly subordinate to the CSAF. A DRU performs a mission that does not fit into any of the MAJCOMs. A DRU has many of the same administrative and organizational responsibilities as a MAJCOM (Example of a DRU: USAF Academy).

DoD Directive 8140.01 (Cyberspace Workforce Management). Unifies the overall cyberspace workforce and establishes specific workforce elements (cyberspace effects, cybersecurity, and cyberspace information technology (IT)) to align, manage and standardize cyberspace work roles, baseline qualifications, and training requirements.

Duty Position Tasks. The tasks assigned to an individual for the position currently held. These include, as a minimum, all core tasks that correspond to the duty position as directed by the AFCFM or MFM, and tasks assigned by the supervisor.

Education and Training Course Announcement (ETCA). Located at https://usaf.dps.mil/teams/app10-etca/SitePages/home.aspx, the ETCA contains specific MAJCOM procedures, fund cite instructions, reporting instructions, and listings for those formal courses the MAJCOMs or FOAs conduct or manage. The ETCA contains courses the Air Force and reserve forces conduct or administer and serves as a reference for the Air Force, DoD, other military services, government agencies, and security assistance programs.

Enlisted Specialty Training (EST). A mix of formal training (technical school) and informal training (on the-job) to qualify and upgrade Airmen in each skill level of a specialty.

Exportable Training. Additional training via computer assisted, paper text, interactive video, or other necessary means to supplement training.

Field Operating Agency (FOA). FOAs are subdivisions of the Air Force directly subordinate to a headquarters US Air Force functional manager. An FOA performs field activities beyond the scope of any of the MAJCOMs. The activities are specialized or associated with an Air Force-wide mission (An example of a FOA is the Air Force Weather Agency).

Field Training. Technical, operator, and other training that either a field training detachment or field training team conducts at operational locations on specific systems and associated direct-support equipment for maintenance and aircrew personnel.

Functional Area Manager (FAM). The individual accountable for the management and oversight of all personnel and equipment within a specific functional area to support the operational planning and execution. Responsibilities include, but are not limited to, developing and reviewing policy; developing, managing, and maintaining Unit Type Codes (UTC); developing criteria for and monitoring readiness reporting; force posturing; and analysis. At each level of responsibility (Headquarters Air Force, MAJCOM, Air Component, FOA, DRU, and Unit), the FAM should be the most highly knowledgeable and experienced person within the functional area and have the widest range of visibility over the functional area readiness and capability issues.

Functional Manager. An individual assigned collateral responsibility for training, classification, utilization, and career development of enlisted personnel. AFSC Functional Managers exist at MAJCOM, NAF and base level.

Go/No-Go. The "Go" is the stage at which a trainee has gained enough skill, knowledge, and experience to perform the tasks without supervision, meeting the task standard. "No-Go" is the stage at which the trainee has not gained enough skill, knowledge, and experience to perform task without supervision, does not meet task standard.

Individual Training Plan (ITP). Using AF Form 623, *On-the-Job Training Record* in conjunction with the approved Air Force training system of record. The AF Form 623 reflects past and current qualifications, and is used to determine training requirements. It is intended to be a complete history of past training and current qualifications. Supervisors will ensure all documentation is accurate and comprehensive.

Initial Skills Training (IST). A formal school course that results in an AFSC 3-skill level award for enlisted or mandatory upgrade training to qualified officers.

Instructional System Development (ISD). A deliberate and orderly (but flexible) process for planning, developing, implementing, and managing instructional systems. It ensures personnel are taught in a cost efficient way to become educated on the knowledge, skills, and abilities essential for successful job performance.

Major Command (MAJCOM). A MAJCOM represents a major Air Force subdivision having a specific portion of the Air Force mission. Each MAJCOM is directly subordinate to HQ USAF. MAJCOMs are interrelated and complementary, providing offensive, defensive, and support elements.

Master Task Listing (MTL). A comprehensive list (100%) of all tasks performed within a work center and consisting of the current CFETP or AFJQS and locally developed AF Forms 797 (as a minimum). Should include tasks required for deployment and/or UTC requirements.

Master Training Plan (MTP). Employs a strategy for ensuring the completion of all work center job requirements by using a MTL and provides milestones for task, CDC completion, and prioritizes deployment/UTC, home station training tasks, upgrade, and qualification tasks.

Occupational Analysis Report (OAR). A detailed report showing the results of an occupational survey of tasks performed within a particular AFSC.

Offensive Cyberspace Operations (OCO). Operations intended to project power by application of force in and through cyberspace (DOD Dictionary of Military and Associated Terms).

On-the-Job Training (OJT). Hands-on, over-the-shoulder training conducted to certify personnel in both upgrade (skill level award) and job qualification (duty position) training.

Proficiency Training. Additional training, either in-residence or exportable advanced training courses, or on-the-job training, provided to personnel to increase their skills and knowledge beyond the minimum required for upgrade.

Qualification Training. Hands-on, task performance based training designed to qualify Airmen in a specific duty position. This training program occurs both during and after the upgrade training process and is designed to provide skills training required to do the job.

Resource Constraints. Resource deficiencies (such as money, facilities, time, manpower, and equipment) that preclude desired training from being delivered.

Specialty Training Requirements Team (STRT). A meeting chaired by the AFCFM with MAJCOM FMs, AETC Training Managers, Subject Matter Experts (SME), and HQ AETC Occupational Analysis Division (OAD) in attendance. Typically held in conjunction with a Utilization and Training Workshop (U&TW) to finalize any CFETP changes or enlisted classification directory descriptions.

Specialty Training Standard (STS). An Air Force publication that describes an Air Force specialty in terms of tasks and knowledge that an Airman in that specialty may be expected to perform or to know on the job. Also identifies the training provided to achieve a 3-, 5-, 7-, or 9-skill level within an enlisted AFS. It further serves as a contract between AETC and the functional user to show which of the overall training requirements for an Air Force Specialty Code (AFSC) are taught in formal schools and correspondence courses.

Standard. An exact value, a physical entity, or an abstract concept established and defined by authority, custom, or common consent to serve as a reference, model, or rule in measuring quantities or qualities, establishing practices or procedures, or evaluating results. It is a fixed quantity or quality.

System Training Plan (STP). A living document that explains what training is needed for a system and how to obtain the training.

Task Module (TM). A group of tasks performed together within an AFS that require common knowledge, skills, and abilities. TMs are identified by an identification code and a statement.

Total Force. All collective components (active, reserve, guard, and civilian elements) of the United States Air Force.

Training Advisory Group (TAG). Chaired by the AFCFM and attended by the MAJCOM, selected DRU and FOA functional managers. The TAG sets training goals and priorities, reviews training programs and evaluates emerging training technologies. The group meets, as required, to prioritize training product development.

Training Capability. The ability of a unit or base to provide training. Authorities consider the availability of equipment, qualified trainers, and study reference materials, and so on in determining a unit's training capability.

Training Planning Team (TPT). Comprised of the same personnel as a U&TW, TPTs are more intimately involved in training development and the range of issues examined is greater than in the U&TW forum.

Training Requirements Analysis (TRA). A detailed analysis of tasks for a particular AFSC to be included in the training decision process.

Training Setting. The type of forum in which training is provided (formal resident school, on-the-job, field training, mobile training team, self-study, etc.).

Unit Type Code (UTC). A five-character alphanumeric code identifying a specific force package of personnel and/or equipment. The UTC is the means for linking logistics and manpower details within a unit type and is used to communicate force data. The UTC represents a wartime capability designed to fill a valid contingency requirement.

Upgrade Training. Training that leads to the award of a higher skill level.

Utilization and Training Workshop (U&TW). A forum of the AFCFM, MAJCOM Functional Managers, subject matter experts (SME), and AETC training personnel that determines career ladder training requirements.

Wartime Tasks. Those tasks that must be taught when courses are accelerated in a wartime environment. In response to a wartime scenario, these tasks will be taught in the 3-level course in a streamlined training environment. These tasks are only for those career fields that still need them applied to their schoolhouse tasks.

Work Center Job Qualification Standard (WCJQS). Work center document that standardizes on-the-job training for Airmen assigned to cyber warfare operations. This document identifies the majority of duties and tasks required to attain qualification. Use this document to plan and record all duty-position related training, and as a basis for preparing a Master Task Listing (MTL) for each functional area. The WCJQS is used by unit training managers, supervisors, trainers, trainees, and other training functions to plan, conduct, and document OJT.

Section A – General Information

1. Purpose of the CFETP. This CFETP provides the information necessary for AFCFMs, MAJCOM Functional Managers (MFM), commanders, training managers, supervisors, trainers and certifiers to plan, develop, manage and conduct an effective and efficient career field training program. The plan outlines the initial skills, upgrade, qualification, advanced and proficiency training that individuals in AFSC 1B4X1 should receive in order to develop and progress throughout their careers. Initial skills training is the AFS specific training an individual receives upon entry into the AF or upon retraining into this specialty for award of the 3-skill level. This training is provided by the 333rd Training Squadron (TRS) at Keesler AFB, MS. Upgrade training identifies the mandatory courses, task qualification requirements, Career Development Course (CDC) completion and correspondence courses required for award of the 5-, 7-, or 9-skill level. Qualification training is actual hands-on task performance training designed to qualify an airman in a specific duty position. This training program occurs both during and after the upgrade training process. It is designed to provide the performance skills and knowledge required to do the job. Advanced training is formal specialty training used for selected airmen. Proficiency training is additional training, either in-residence or exportable advanced training courses, or on-the-job training provided to personnel to increase their skills and knowledge beyond the minimum required for upgrade. The CFETP has several purposes, some of which are:

1.1. Serves as a management tool to plan, develop, manage, and conduct a career field training program. Also, ensures that established training is provided at the appropriate point in an individual's career.

1.2. Identifies task and knowledge training requirements for each skill level in the specialty and recommends training throughout each phase of an individual's career.

1.3. Lists training courses available in the specialty, identifies sources of the training, and provides the training medium.

1.4. Identifies major resource constraints that impact implementation of the desired career field training program.

2. Use of the CFETP. The CFETP is maintained by the 1BXXX AFCFM, HAF A6/OF. MAJCOM FMs and AETC review the plan annually to ensure currency and accuracy and forward recommended changes to the AFCFM. Using the list of courses in Part II, they determine whether duplicate training exists and take steps to eliminate/prevent duplicate efforts. Career field training managers at all levels use the plan to ensure a comprehensive and cohesive training program is available for each individual in the career ladder.

2.1. AETC training personnel develop/revise formal resident and exportable training based upon requirements established by the users and documented in the STS. They also develop procurement and acquisition strategies for obtaining resources needed to provide the identified training.

2.2. MAJCOM FMs ensure their training programs complement the CFETP mandatory initial skill and upgrade requirements. They also identify the needed AFJQSs/AFQTPs to document unique upgrade and continuation training requirements. Requirements are satisfied through OJT, resident training, contract training, or exportable courseware/courses. MAJCOM developed training to support this AFSC must be included into this plan.

2.3. 81 TRSS/TSQ Qualification Training Flight (Q-Flight) personnel develop training packages (AFJQSs/AFQTPs) based on requests submitted by the MAJCOMs and according to the priorities assigned by the AFCFM.

2.4. Unit training managers and supervisors manage and control progression through the career field by ensuring individuals complete the mandatory training requirements for upgrade specified in this plan and supplemented by their MAJCOM. The list of courses in Part II is used as a reference for planning continuation or career enhancement training.

2.5. Submit recommended CFETP corrections, additions, and deletions through your MAJCOM Functional Manager.

3. Coordination and Approval of the CFETP. The AFCFM is the approval authority. MAJCOM representatives and AETC training personnel coordinate on the career field training requirements. The AETC training manager initiates an annual review of this document by AETC and MAJCOM functional managers to ensure the CFETP's currency and accuracy by using the list of courses in Part II to eliminate duplicate training.

Section B - Career Field Progression ad Information

4. Specialty Description. Performs duties to develop, sustain, and enhance cyberspace capabilities to defend national interests from attack and to create effects in cyberspace to achieve national objectives. Plans and conducts Offensive Cyberspace Operations (OCO), Defensive Cyberspace Operations (DCO), Cyber Network Operations (CNO), Cyber Network Defense (CND), and cryptologic activities using established tactics, techniques and procedures (TTPs) to achieve Service, CCMD, and national objectives**.** Executes command and control (C2) of assigned cyberspace forces and de-conflicts cyberspace operations across the kinetic and non-kinetic spectrum. Supports cyberspace capability development, testing and implementation. Partners with Joint, Interagency, Intergovernmental, and Multinational forces to detect, deny, degrade, disrupt, destroy, manipulate, and mitigate adversarial access to sovereign national or partner cyberspace systems. Related DoD Occupational Subgroup: 153100

5. Skills and Career Progression.

5.1. Cyber Warfare Operations Apprentice (1B431). Initial skills in this specialty consist of the tasks and knowledge provided in the 3-skill level resident 1B4X1 Cyber Warfare Operations Apprentice Course located at Keesler AFB MS. Individuals must complete the initial skills course(s) to be awarded AFSC 1B431. Current requirements were identified and validated during the STRT held 23 – 27 Feb 2026.

5.2. Cyber Warfare Operations Journeyman (1B451). To qualify for the 5-skill level, Airmen must: (1) complete the 5-level Career Development Course(s) (CDC), if applicable; (2) meet mandatory requirements listed in the specialty description in the Department of the Air Force Enlisted Classification Directory (DAFECD) and CFETP. Once qualifications are met, Airmen must be upgraded unless not recommended by their commander. Supervisors may identify and standardize local tasks for UGT with the AFCFM approval. Coordinate requests for AFCFM approval through the MAJCOM FM. UGT consists of completing duty position training and/or certification, any specified core task training, and appropriate courses as outlined in the WCJQS.

5.3. Cyber Warfare Operations Craftsman (1B471). To qualify for award of the 7-level, Airmen must (1) be a SSgt or higher (SrA with line number for SSgt will be entered into 7-skill level upgrade training); (2) complete CDCs, if applicable (3) meet mandatory requirements listed in the specialty description in the AFECD and CFETP. Once qualifications are met, Airmen must be upgraded unless not recommended by their commander. Individuals in retraining status are subject to the same requirements. Supervisors may identify and standardize local tasks for UGT with the AFCFM approval. Coordinate requests for AFCFM approval through the MAJCOM FM.

5.4. Cyber Warfare Operations Superintendent (1B491). SMSgts must upgrade upon promotion notification and completion of mandatory requirements listed in the AFECD.

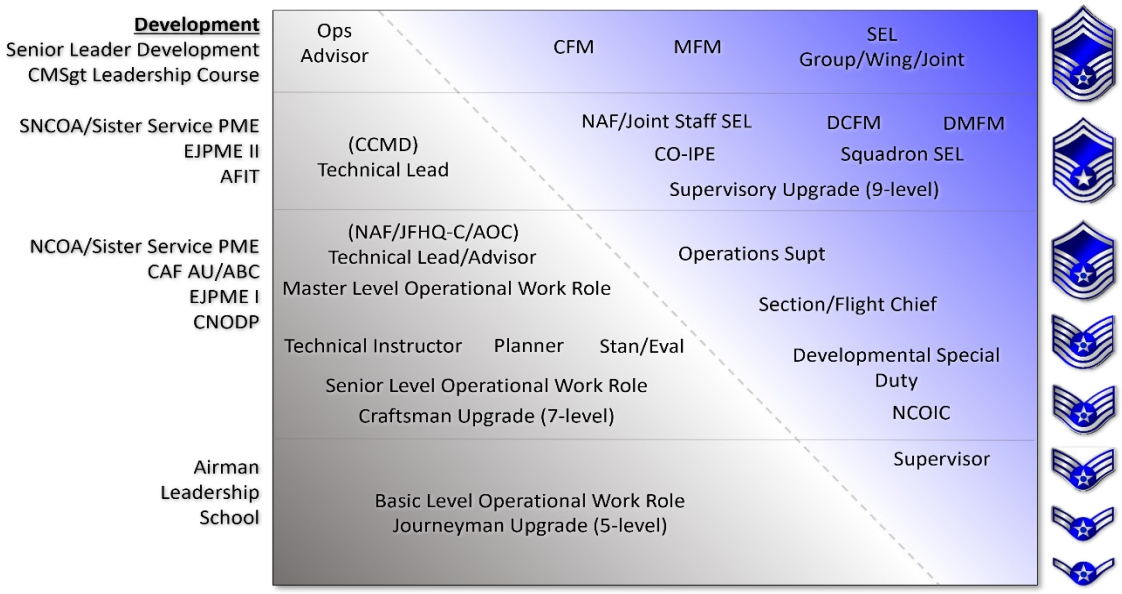
5.5. Occupational Badges. The Cyberspace Operations Badge is the Cyber Warfare Operations occupational badge and is authorized to be worn by 1BXXX Airmen. See DAFI 36-2903, *Dress and Personal Appearance of Air Force Personnel* for proper wear guidance. The below guidance outlines requirements for the wear of the Basic, Senior, and Master Cyberspace Operations badges.

5.5.1. Basic Cyberspace Operations Badge: This badge will be worn upon graduation of the Cyber Warfare Operations Initial Skills Training Course.

5.5.2. Senior Cyberspace Operations Badge: This badge will be worn after award of 7-skill level.

5.5.3. Master Cyberspace Operations Badge: This badge will be worn upon award of 9-skill level or worn as a Master Sergeant with 5 years of experience within the 1BXXX, Cyber Warfare Operations career field.

1BXXX Career Path Chart



5.6. Enlisted Development Team (EDT).

5.6.1. Mission: The Cyber Warfare Operations development team provides oversight of enlisted development within the 1B career field to meet both functional and Air Force corporate leadership requirements. The EDT outlines the training, education, and experience requirements for critical Cyber Warfare Operations duty positions, and provides vector recommendations for qualified Senior Non-Commissioned Officers (SNCOs) into Key Leadership Positions (KLPs) across the Air Force. The EDT also recommends other developmental opportunities for Cyber Warfare Operations SNCOs to facilitate deliberate development. Recommendations and vectors are reflective of the EDT's collective board process and are based on the member's experience. The EDT process will highlight positional opportunities that a member should be considered for and seek out for professional growth. Vectoring will consist of recommendations for identified positions within the Cyber Warfare Operations construct for which a member should be considered in subsequent assignments but will not identify specific assignment locations.

5.6.2. Process: The functional manager chairs the EDT (unless delegated) and determines the membership. The career field manager may chair the EDT on behalf of the functional manager. EDT panel members are the Cyber Warfare Operations Career Field Manager, major command-level functional leadership, and other career field strategic-level subject matter experts at the Senior Master Sergeant (SMSgt) and/or Chief Master Sergeant (CMSgt) levels as determined by the chair. Members meeting the EDT can express their own desires through the Individual Development Plan (IDP) for consideration by the board. During the EDT review process, EDT panel members assess billets for key leadership or key development position designation. EDT panel members will also board and evaluate each 1Bs past performance, duty history and experience, scope of responsibility, decorations, and IDP input to determine appropriate development and vector recommendations. Once complete, 1Bs will be made aware of their vector/tier through myVector.

6. Training Decisions.

6.1. Three-Skill Level Course. The 3-skill level course is overhauled to continue the evolution of the Cyber Warfare Operations career field. The CFM and MAJCOM Functional Managers concur on the course changes.

6.2. Five-Skill Level Upgrade Requirements. 5-level core task requirements as well as CDC requirements have been adjusted to meet the needs of the operational community.

6.3. Seven-Skill Level Upgrade Requirements. 7-level core task requirements as well as CDC requirements have been adjusted to meet the needs of the operational community.

6.4. Nine-Skill Level Upgrade Requirements. 9-level core task requirements as well as CDC requirements have been adjusted to meet the needs of the operational community.

6.5. Occupational Competency Model (OCM). The 1BXXX AFSC transitioned to the OCM in order to create a scalable, standardized, and repeatable framework for developing Airmen across their entire careers, rather than just in initial training. This model aims to reduce the decay of perishable skills, improve career-long proficiency, and align talent management with evolving operational needs, such as the Multi-Capable Airman initiative.

6.6. Proficiency Training. Any additional knowledge and skills that were not provided through initial skills or upgrade training fall under the auspices of continuation training. The purpose of the continuation program is to provide additional training that exceeds minimum qualification or upgrade training requirements with an emphasis on present and future duty positions. MAJCOMs and joint activities must develop a continuation-training program that ensures personnel in the Cyber Warfare Operations Specialty receive the necessary training at the appropriate point in their career. The training program will identify both mandatory and optional training requirements. The below describes examples of training course opportunities that exist within the joint community and pertain to cyberspace application; it is not an exhaustive list and other opportunities may be found. The goal of capturing some of these courses is to expand awareness of existing enhancement opportunities.

6.6.1. Joint Targeting School (JTS). This school is located at Dam Neck Naval Air Station, VA and hosts four intermediate level courses. The Joint Targeting Staff Course focuses on the application of the six-step Joint Targeting Cycle at the strategic and operational levels of war and involves the presentation of concepts and theory associated with each step. The Joint Targeting Applications Course focuses on the Weaponeering process and concepts of weapon delivery accuracy, damage mechanisms, and damage criteria along with an introduction to the software tools used for damage prediction calculations. The Joint Battle Damage Assessment (BDA) Course is focused on the methodologies employed to accurately assess and communicate the effectiveness of military force delivered against a variety of generic targets and target models. The Joint Collateral Damage Estimation Course is based on Chairman of the Joint Chiefs of Staff Manual (CJCSM) 3160.01 and focuses on the CDE Assessment Process, Casualty Estimation, Mitigation Techniques, and CDE Automation Tools.

6.6.2. Joint Intermediate Target Development Course (JITD Course). The goal of the Joint Intermediate Target Development Course is to ensure targeting analysts have the skills to develop and database any target type according to the standards directed in CJCSI 3370.01A, Target Development Standards. Work-roles who utilize this course include planners and analysts who learn to database Electronic Targeting Folder (ETF) remarks from facilities, individuals, virtual entity, equipment, and organizational (FIVE-O) target types.

6.6.3. Joint Network Attack Course (JNAC). The focus of this course is to educate staff level planners on how operations are conducted across the spectrum of Cyberspace Operations, focusing on Offensive Cyberspace Operations (OCO) and Defensive Cyberspace Operations (DCO). Additionally, students are trained in the various skills to conduct OCO/DCO planning and develop an understanding of technical versus operational terminology, as well as domestic and international laws as they apply to OCO/DCO.

6.6.4. Joint Cyberspace Operations Planners Course (JCOPC). This course is sponsored by the USSTRATCOM Cyberspace Training Initiative (CTI) and managed by USCYBERCOM. JCOPC was originally developed as a mobile training curriculum to be presented to Combatant Commands (CCMDs) in response to identified training requirements. Its purpose is to training CCMD and Joint Force Command (JFC) staffs on the integration of cyberspace operations into CCMD/JFC level planning. Its scope has been expended to be included as part of the training pipeline for the Cyber Mission Force (CMF).

6.6.5. Joint Information Operations Planners Course (JIOPC). This course is an informative and practical course of instruction that is particularly useful for personnel who are headed to Information Operations (IO) planning jobs at the operational level of war (CCMD/JTF/JFC). The course is composed of one week of lectures on the IO core, supporting and related capabilities specified in Joint IO doctrine, followed by three weeks of lectures and practical exercises on the Joint Planning process and how to use it for IO planning.

6.6.6. Joint Advanced Cyber Warfare Course (JACWC). This course is designed to make personnel, regardless of background, effective and credible within the cyber warfare community upon graduation. JACWC provides and orientation to USCYBERCOM, the global cryptologic platform, the Intelligence Community, the US Government cyber community of interest, allies, and major partners in the conduct of cyber warfare, as well as cyber warfare threats, operations, planning, and analysis of desired effects.

6.6.7. Air Operations Center Courses. The 505th Training Squadron at Hurlburt Field, FL teaches 16 Initial Qualification Training courses for the AOC weapons system. Additionally, the squadron teaches the joint certified Joint Air Operations Center Command and Control Course (JAOC2C) to students in order to cover the theater air planning process.

6.6.8. Air Force Forces (AFFOR) Intermediate Staff Course (AISC). This is a five-day course which trains Component-Numbered Air Force (C-NAF) Forces staff personnel who conduct and support operational-level planning and execution. The course develops the foundational skills necessary to execute warfighting headquarters staff responsibilities on behalf of the Commander Air Force Forces (COMAFFOR) as part of the C-NAF in support of Joint Force Commander objectives. This course is run through the 505th Combat Training Squadron at Hurlburt Field, FL.

6.6.9. Joint C4I Cyber Course. The JC4I Cyber Course prepares students for positions that require an understanding of command, control, communications, computers, and intelligence/cyber (C4I/C). This course educates personnel from varied backgrounds to function in a joint environment. It provides a broad understanding of joint C4I/C doctrine and current policy guidance. Students apply joint C4I/C concepts and skills/procedures to prepare for duty in joint or service C4I/C staff operations and planning assignments. The program covers a wide spectrum of C4I/C that extends from the national and strategic levels to the theater and tactical levels in support of the President, Secretary of Defense, and commanders who control military forces. This course is geared at intermediate level staff officers, senior noncommissioned officers, and DoD civilian equivalents for duty in C4I/C staff operations at a Combatant Command, Joint Task Force, or Headquarters staff. The course is open to grades E-6 to E-9 and is 3-weeks long, located at the Joint Forces Staff College in Norfolk, VA.

6.6.10. Cyberspace Tactical Planner Course. This course was developed by weapons officers at the 67th Cyberspace Wing to teach large force employment, roles and responsibilities of the mission commander and tactical planners, and integration of cyberspace forces identified in Air Force Cyberspace Tasking Orders. The training program teaches Air Force Weapons School ME3C-$(PC)^2$ planning process methodology for both offensive and defensive mission sets.

6.6.11. Enlisted Joint Professional Military Education (EJPME). Both EJPME I and II courses are stand-alone, 100% online, web-based courses that uses multi-media instruction. Course eligibility is restricted to E-5 and above for EJPME I and E-7 and above for EJPME II. The EJPME courses prepare senior enlisted leaders assigned to joint organizations to successfully support activities and supervise multiple Service members. Upon completion of the course, SELs will be more competent, confident, and prepare to assimilate and effectively contribute in joint assignments as well as mentor junior enlisted leaders and other service personnel they supervise. These courses may be found on Joint Knowledge Online (JKO).

6.7. Degree Programs:

6.7.1. Air Force Institute of Technology (AFIT)

6.7.1.1. Mission: The Air Force Institute of Technology, or AFIT, is the Air Force's graduate school of engineering and management as well as its institution for technical professional continuing education. A component of Air University and Air Education and Training Command, AFIT is committed to providing defense-focused graduate and professional continuing education and research to sustain the technological supremacy of America's air and space forces.

6.7.2. National Intelligence University (NIU)

6.7.2.1. Description: The National Intelligence University (NIU) is a regionally accredited institution offering military and civilian personnel working in intelligence or intelligene/security-related specialties the opportunity to enroll in professional intelligence undergraduate or graduate-level sudy in full- or part-time programs. Coursework concentrates on a variety of intelligence disciplines to include collection; analysis; regional studies; information operations; intelligence, surveillance, and reconnaissance; national security issues; and strategic intelligence in the joint environment. NIU educates future leaders who are full partners with their policy, planning, and operations counterparts and who are able to anticipate and tailor the intelligence required at the national, theater and tactical levels. Classes have an Intelligence Community student mix from all services and the federal government. The college is located at Roberdeau Hall on the IC Campus Bethesda (ICC-B) in Bethesda, MD.

6.7.2.1.1. Coursework Background: In 2012, the School of Science and Technology Intelligence was approved by the Department of Education to offer a Master of Science and Technology Intelligence (MSTI) Degree to individuals in both the cyber and intelligence career fields. The MSTI is a graduate degree program requiring complation of an 11-month course of study consisting of five required courses, seven electives, and thesis courses. 1D7/1B4 applicants will adhere to the Cyber Intelligence and Data Analytics Concentration of the MSTI program, which will include study in areas such as social networks and intelligence, foreign cyber strategies, and information influence and deception.

6.7.2.2. Eligibility Criteria:

6.7.2.2.1. NIU programs are open to military service members and U.S. government employees who are U.S. citizens and who hold finalized Top Secret/SCI clearances.

6.7.2.2.2. The program is open to Air Force activity duty and Reserve Component NCOs in the grades of E-5 select through E-8 and civilians from GG-09 to GG-15.

6.7.2.2.3. Military applicants must be PCS eligible. Further criteria are defined annually and conveyed via an AF message to the field.

6.7.2.2.4. Military members must have three years retainability upon class graduation. Personnel will incur a three-year active duty service commitment upon graduation/program completion.

6.7.2.2.5. Military members must have completed Professional Military Education commensurate with their grade.

6.7.2.2.6. Baccalaureate Degree for a regionally accredited institution and the graduate record exam (GRE) for those who do not already have a Master's degree from a regionally accredited institution. The ideal candidate will have academic exposure to science, technology, engineering or math (STEM) during their undergraduate years.

6.7.2.2.7. Other eligibility requirements apply. See call for nominations guidance or below Application Process.

6.7.2.2.8. Application Process: Air Staff calls for nominations for this program annually in the summer to fall timeframe via formal message traffic. For further information on applying to the National Intelligence University, visit http://www.ni-u.edu/

6.8. Computer Network Operations Development Program (CNODP).

6.8.1. Program Overview: CNODP is technically demanding three-year program with a follow-on assignment individually tailored to each applicant to best capitalize on his/her expertise. This is an opportunity for a select group of highly technically-inclined individuals to further develop their skills in the areas of secure system design, vulnerability analysis, computer network defense (CND), and computer network exploitation (CNE). The goal of the program is to develop a cadre of technical leaders who will improve the Department of Defense's and each military service's Computer Network Operations (CNO) capabilities. The CNODP program is made up of CNODP, the National Security Agency's (NSA) program for civilians that includes 10 or more slots annually for military members, and USCYBERCOM's Joint Cyber Development Program (JCDP), which allocates 20 slots for military members. All 30 slots are competitive across the services.

6.8.2. Additional Information: CNODP begins with three to four months of "core training", which consists of both internal and external classes to enhance the intern's technical skills and bridge gaps between typical Computer Science/Engineering curriculum and those necessary for Computer Network Attack / Exploitation / Defense. Following core training, CNODP interns tour for six to twelve months each in various offices at NSA and USCYBERCOM, working as a CNO developer, operator, and/or analyst. The total time spent touring offices is 30 months, resulting in three to five tours for each intern; however, interns choose the tours they want to work. Each intern must complete at least one offensive and at least one defensive tour during the program. These tours are not limited to the primary NSA campus but most will be within 30 miles of Fort Meade. JCDP participants must also complete one tour in support of USCYBERCOM. The final three months of the program are spent working one of two team "final projects", which are hard CNO problems that NSA and USCYBERCOM need solved. More information can be found here: https://www.intelligencecareers.gov/nsa/development-program

7. Community College of the Air Force (CCAF) Academic Programs. Enrollment in CCAF occurs upon completion of basic military training. CCAF provides the opportunity for all enlisted members to obtain an Associate in Applied Science degree. Refer to the AF Virtual Education Center (accessible via the AF Portal, https://www.my.af.mil) for CCAF credits earned for technical training courses attended. In order to be awarded a CCAF AAS degree, students must complete the program before they separate from the Air Force, retire, or are commissioned as an officer.

7.1. The Cybersecurity (0CYC) program applies to the 1B4X1 career field.

7.1.1. Degree Requirements: Individuals must hold the 5-skill level at the time of program completion.

	Semester Hours
Technical Education	24
Leadership, Management, and Military Studies	6
General Education	15
Program Electives	15
Total	60

7.1.2. Technical Education (24 semester hours): A minimum of 9 semester hours of technical core subjects and courses must be applied and the remaining semester hours will be applied from technical core/technical elective subjects and courses. Requests to substitute comparable courses or to exceed specified semester hour values in any subject/course must be approved in advance by the technical branch of the CCAF Administrative Center.

7.1.3. Leadership, Management, and Military Studies (LMMS) (6 semester hours): Professional military education (PME) and/or civilian management courses accepted in transfer and/or by testing credit. See CCAF General Catalog for application of civilian management courses.

7.1.4. General Education (15 semester hours): Courses must meet the criteria for application of courses to the General Education requirement and be in agreement with the definitions of applicable General Education subjects/courses as outlined in the CCAF General Catalog.

7.1.5. General Education Mobile (GEM): GEM is a partnership between CCAF and civilian academic institutions to offer general education courses to meet CCAF A.A.S. degree requirements. Courses are offered via distance learning which reduces CCAF educational impact of deployments, PCS and family commitments.

7.1.6 Program Elective (15 semester hours): Courses applying to technical education, LMMS or general education requirements; natural science courses meeting general education requirement application criteria; foreign language credit earned at Defense Language Institute or through the Defense Language Proficiency Test; maximum 9 Semester Hours of CCAF degree-applicable technical course credit otherwise not applicable to program of enrollment.

7.2. See the current CCAF General Catalog for details regarding the Associates of Applied Science in Cybersecurity. The catalog is available at your education office or online at https://www.airuniversity.af.edu/CCAF/.

7.3. Additional off-duty education is a personal choice that is encouraged for all. Individuals desiring to become an AETC Instructor must possess as a minimum an associate degree or should be actively pursuing an associate degree. Special Duty Assignment (SDA) requires an AETC instructor candidate to have a CCAF degree or be within one year of completion (45 semester hours). A degreed faculty is necessary to maintain accreditation through the Southern Association of Colleges and Schools.

8. Career Field Path. Table 8.1. identifies career milestones for the 1B4X1 Air Force specialty.

Table 8.1.

1B4X1 CYBER WARFARE OPERATIONS CAREER PATH			
	GRADE REQUIREMENTS		
Education and Training Requirements	***Rank***	***Earliest Sew-On***	***High Year Of Tenure (HYT)***
Basic Military Training School (BMTS)			
Apprentice Technical School (3-Skill Level)	AB		
Upgrade To Journeyman (5-Skill Level) - Specific AFJQSs/AFQTPs for equipment at assigned location by duty position. (see NOTE 2) - Complete CDC if applicable	A1C SrA	28 months	10 Years
Airman Leadership School - Must be a SrA with 48 months time in service or be a SSgt Selectee. - Resident graduation is a prerequisite for SSgt sew-on (Active Duty Only).	**Trainer** - Qualified and certified to perform the task to be trained. - Must attend formal DAF Training Course. - Recommended by the supervisor.		
Upgrade To Craftsman (7-Skill Level) - Minimum rank of SSgt or SSgt Selectee. - Complete CDC if applicable. - Specific AFJQSs/AFQTPs for equipment at assigned location by duty position.	SSgt (Sel)	3 years	20 years

1B4X1 CYBER WARFARE OPERATIONS CAREER PATH			
	GRADE REQUIREMENTS		
Education and Training Requirements	***Rank***	***Earliest Sew-On***	***High Year Of Tenure (HYT)***
Non-Commisioned Officer Academy - Must be a TSgt or TSgt Selectee. - Resident graduation is a prequisite for MSgt sew-on (Active Duty Only).	TSgt	5 years	22 years
USAF Senior NCO Academy - Must be a SMSgt or SMSgt Selectee. - Resident graduation is a prerequisite for SMSgt sew-on (Active Duty Only). - Guard and Reserve who have completed NCOA may enroll in SNCOA DLC.	MSgt	8 years	24 years
Upgrade To Superintendent (9-Skill Level) **MANDATORY** - Minimum rank of SMSgt.	SMSgt (Sel)	11 years	26 Years
Chief Enlisted Manager (CEM)	CMSgt	14 years	30 years
NOTE 1: Published sew-on times are Air Force averages. Refer to the myFSS website for current information: https://myfss.us.af.mil/. **NOTE 2:** See Part II, Sections C and D for a list of AFJQSs/AFQTPs and AETC supplemental training.			

Section C - Skill Level Training Requirements

9. Purpose. Skill-level training requirements in this specialty are defined in terms of tasks and knowledge requirements. This section outlines the specialty qualification requirements for each skill-level in broad, general terms and establishes the mandatory requirements for entry, award, and retention of each skill-level. The specific task and knowledge training requirements are identified in the STS at Part 2, Section A and B of this CFETP.

10. Specialty Qualification Requirements. This information supplements the DAFECD. AFS-specific specialty qualifications can be found in the DAFECD.

11. Competencies. The Air Force defines competencies as a combination of knowledge, skills, abilities, and other characteristics that manifest in observable and measurable patterns of behaviors required for mission success. The Air Force has rebuilt the entire training platform by using competencies to identify the behaviors that are needed to be successful on the job. This has been accomplished by identifying and integrating the 1BXXX, Cyber Warfare Operations occupational competencies and the Air Force's foundational competencies in a manner that provides all Airmen with transparent and unbiased pathways towards their own successful development.

11.1. Airman's Foundational Competencies. The foundational competencies are a set of accepted and valued competencies, which enable success across a wide array of DAF missions, roles, functions, and duties. These competencies are the core of Airmen development and enable Airmen with tools, pathways, and capabilities to improve their performance in any job, specialty, or situation. The foundational competencies are grouped into different categories of Developing Self, Developing Others, Developing Ideas, and Developing Organization. Airmen can go to MyVector (accessible via AF Portal) to complete a self-assessment, which will have them evaluate themselves on the 23 Airman's Foundational Competencies or a 360-degree assessment, where subordinates, peers, and leaders can also provide feedback. The assessment tools will provide Airmen with immediate feedback on personal strengths and areas for improvement. Additionally, a personal improvement plan with targeted resources (videos, reading content, developmental opportunities) are provided for continued self-development. Airman's Foundational Competencies can be found in AFH 36-2647.

11.2. Occupational Competencies. Occupational competencies are a set of competencies required of all Airmen within a specific workforce category. These competencies provide a framework that describes the knowledge, skills, abilities, and other characteristics needed to perform that function's mission successfully.

11.2.1. Occupational Competency Model. A career field's competencies can be viewed in a competency model, which is an organized collection of competencies pertinent to the career field. The occupational competency model provides a framework to effectively assess, maintain, and monitor the competencies required for mission success for Airmen within the Cyber Warfare Operations community. The occupational competency modeling process follows a distinct process with continued involvement from the career field and allows Airmen to see how their task lists, OJT, formal courses, and other training, education, and experiences are aligned to the career field's strategic objectives.

11.2.2. Career fields work with trained competency experts to identify and develop their competency model, which consists of the competencies, sub-competencies, and definitions. Occupational competency models will be different for each career field. The model focuses on integrating not just the technical components, but also leadership, management, combat, joint, all-domain, and social mastery competencies required for Airmen to succeed in their career field.

Competency	Sub-Competency	Description
Operations	Planning	The practice of translating guidance into actionable tasks by balancing stakeholder needs and risks in order to achieve mission objectives.
	Communication	The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.
	Leadership	The ability to guide and influence individuals or groups toward achieving a common goal, encompassing skills like direction, motivation, and decision-making by fostering collaboration, accountability, innovation, technical expertise, and personnel well-being.
	Readiness	The state of preparedness to respond swiftly and effectively to meet mission requirements.
	Training	The process of developing Airmen with the knowledge, skills, and abilities necessary for effective mission performance.
Airmanship	Personnel Management	The management of people through deliberate planning, institutional training, development, direction, and organization to maximize performance.
	Program Management	The development, implementation, and validation of administrative programs to enhance and sustain organizational performance.
	Resource Management	The allocation, utilization, and oversight of resources to ensure mission readiness and operational effectiveness.
Mission Execution	Electromagnetic Spectrum	The aggregate of knowledge and application of technologies employing electromagnetic radiation, signals and transmissions, tradecraft, and production of effects.
	Hardware	The aggregate of knowledge and application of technologies employing tangible components and firmware, defining infrastructure across physical and virtualized levels.
	Networking	The aggregate of knowledge and technologies employing the logical interconnection of devices to enable information transfer between systems.
	Software	The aggregate of knowledge and technologies to develop the logical instruction sets, data, or programs that tell a system what to do and how to perform specific tasks.
	Analysis	The use of methods and technologies to conduct data analysis (collect, clean, compile, visualize, present, evaluate) to produce information.

Figure 1. 1BXXX, Occupational Competency Model.

11.2.3. Occupational Competency Rubric. After a model is developed, a team of subject matter experts begin building competency rubrics, which consists of a competency, a description of the competency, proficiency levels, and measurable and observable behaviors. The competency rubrics will help Airmen learn which behaviors are aligned to the career field's strategic direction, the professional developmental expectations, and the criteria for success. Figure 2 provides an example of a competency rubric for the (1BXXX) career field.

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	1. Authors and employs strategies (e.g. CONOPS) to address the unique needs of stakeholders 2. Conveys information in actionable language to support mission objectives 3. Facilitates collaboration between stakeholders to integrate and achieve outcomes
Sub-Competencies Communication	Advanced Consistency of Application: Sustained application of competency over time in complex situations	1. Compiles information from various sources to shape operational activities 2. Simplifies technical information into clear and concise language to ensure comprehension 3. Distills guidance into actionable language to support mission objectives
Description The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	1. Develops clear and concise communications to inform stakeholders 2. Utilizes active listening to achieve a shared understanding between communicating parties 3. Distills guidance into simple and actionable language to support mission objectives
Supporting Competencies Analytical Thinking Communication Influence Relationship Building	Basic Consistency of Application: Sustained application of competency over time	1. Transmit or receive information to enable understanding or redistribution 2. Leverages presentation tools to build and brief ideas 3. Seeks additional information to eliminate misunderstanding or clarify intent

Figure 2. 1BXXX Occupational Competency Rubric for Communication.

11.2.4. To better understand how to read and utilize the competency rubric, a breakdown of each component is explained below in figure 2a-c.

Competency Operations	←	The competency section states the competency group.
Sub-Competencies Communication	←	The sub-competency section states the narrower category that forms part of the competency group. Note: Some models may only consist of a competency and not include a sub-competency.
Description The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.	←	The description section provides a statement that gives details about the sub-competency, enabling career field members to better understand how sub-competency relates to the AFS.
Supporting Competencies Analytical Thinking Communication Influence Relationship Building	←	The supporting competencies section are supported-level competencies that are linked to the success of the sub-competency. These competencies lend themselves more toward areas like values, traits, and attitudes. These competencies were included as part of a larger survey that went out to the entire AFS; respondents were asked to rate the top supporting competencies they believe will attribute to higher successful performance within the subcompetency.

Figure 2a. 1BXXX Occupational Competency Rubric for Communication.

Proficiency Levels		The proficiency levels are broken into four parts: basic, intermediate, advanced, and expert. Under each proficiency level are predetermined criteria selected by a group of SMEs from your career field and validated by the career field. The criteria were used as the basis to develop observable behaviors. These criteria provide concrete parameters for the behaviors, which are consistent but progressive in nature as a member moves up the scale from basic to expert. Some of the criteria (e.g. depth of knowledge, consistency of application/complexity, and thinking challenge) allow an individual to become an expert through the experience gained in a particular job and over a period of time. For example, the person can quickly move up different proficiency levels while they are serving as a technician at a flight; they move quickly because they are exposed to a variety of situations. While other criteria (e.g. scope, impact, and reach of influence) require more of a hierarchical approach to gain the experience needed to progress through the competency levels. Moving through the proficiency levels may be difficult to do in certain jobs. For example, if scope at the expert level requires job integration with the AF-level, then the individual may have to be in a position where they can gain that experience (i.e. at HHQ, Wing, or an organization with far reaching capabilities).
Advanced **Consistency of Application:** Sustained application of competency over time in complex situations		
Intermediate **Consistency of Application:** Sustained application of competency over time in a variety of situations		
Basic **Consistency of Application:** Sustained application of competency over time		

Figure 2b. 1BXXX Occupational Competency Proficiency Levels.

Observable Behaviors	The observable behaviors are statements of what can be observed from an individual manifesting the competency at the respective competency level. They provide objective evidence that the individual possesses the competency level, and shows what effective performance looks like. The behaviors are written to be specific enough so they can be observable and lend themselves towards measurement.
1. Authors and employs strategies (e.g. CONOPS) to address the unique needs of stakeholders 2. Conveys information in actionable language to support mission objectives 3. Facilitates collaboration between stakeholders to integrate and achieve outcomes	
1. Compiles information from various sources to shape operational activities 2. Simplifies technical information into clear and concise language to ensure comprehension 3. Distills guidance into actionable language to support mission objectives	
1. Develops clear and concise communications to inform stakeholders 2. Utilizes active listening to achieve a shared understanding between communicating parties 3. Distills guidance into simple and actionable language to support mission objectives	
1. Transmit or receive information to enable understanding or redistribution 2. Leverages presentation tools to build and brief ideas 3. Seeks additional information to eliminate misunderstanding or clarify intent	

Figure 2c. 1BXXX Occupational Competency Observable Behaviors.

11.3. Another key component within the rubric is the supporting competencies section. These are the top four supporting competencies that can help members excel and be successful in that particular sub-competency. Some of these supporting competencies are tied directly to the Airman's Foundational Competencies, while others may be unique to the career field. Having these supporting competencies identified and linked to a career field's competency model can cultivate those underlying characteristics needed to succeed on the job. Leaders, supervisors, trainers, instructors, or mentors can now set members up for greater success by building these supporting competencies and placing their Airmen in situations where they can apply those

strategies. All of these elements come together to ensure we can develop Airmen who are better prepared, present and future mission focused, and ready to succeed in any situation. Additionally, DAFMAN 36-2643, *Coaching and Mentoring Program*, has information on how competencies can be used when an established mentoring strategy is put into effect to foster and develop Airmen.

11.4. Competency Development. The intent of moving towards a competency-based system is to sharpen our Airmen's tactical expertise, operational competence, strategic vision, and joint proficiency to lead and execute the full spectrum of USAF missions. This occurs not in a classroom but on the job by combining education, training, and experiences to provide Airmen with a better developmental pathway as they move along their careers. Airmen are still required to complete specific training courses, core tasks, and other training requirements in order to attain a 3-, 5-, and 7-skill levels. Competency development allows Airmen to move beyond the minimum career field requirements and begin addressing developmental gaps and strengthening their capabilities. The information included within the competency model will allow members within the community to manage their professional growth and development by identifying their strengths and weaknesses against clear and objective behaviors within the competency model.

11.4.1. Below are the competency rubrics for the 1BXXX, Cyber Warfare Operations specialty.

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Scope: Integration with AF-level/within industry	1. Facilitates collaboration across multiple domains to provide situational awareness of the information environment (e.g. linkages outside of the respective AOR) 2. Advises senior leaders of operational environment to inform them of strategic risk 3. Formulates alternative methods to achieve desired effects 4. Manages strategic-level planning effort(s) that spans competition, crisis, and conflict to create executable plan(s)
Sub-Competencies	Advanced Scope: Integration with organizational strategies	1. Develops inputs/outputs at every step of the Joint Planning Process (JPP) to achieve operational success 2. Coordinates with stakeholders on operational goals to balances constraints and restraints 3. Guides executable plan development by means of a standard framework adapting to changes in the environment 4. Identifies alternatives to achieve desired effects and phasing
Planning		
Description	Intermediate Scope: Integration with concerned areas	1. Interprets guidance, authoritative documents, and builds planning deliverables to support desired end-state 2. Integrates intelligence resources and utilize RFI process to validate assumptions 3. Participates in mission planning refinement to highlight gaps and vulnerabilities (e.g. wargaming) 4. Prepare planning products for approval (COA)
The practice of translating guidance into actionable tasks by balancing stakeholder needs and risks in order to achieve mission objectives.		
Supporting Competencies	Basic Scope: Specific Area	1. Identifies direction, guidance, and authoritative documents to inform planning 2. Determines intelligence resources to obtain threat analysis, identify limitations, and assess the environment 3. Determines employment of capabilities to advise COA development through approval 4. Explains the Joint Planning Process steps to facilitate future planning
Analytical Thinking Communication Decision Making Strategic Thinking		

Figure 3. 1BXXX Planning Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	1. Authors and employs strategies (e.g. CONOPS) to address the unique needs of stakeholders 2. Conveys information in actionable language to support mission objectives 3. Facilitates collaboration between stakeholders to integrate and achieve outcomes
Sub-Competencies	Advanced Consistency of Application: Sustained application of competency over time in complex situations	1. Compiles information from various sources to shape operational activities 2. Simplifies technical information into clear and concise language to ensure comprehension 3. Distills guidance into actionable language to support mission objectives
Communication		
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	1. Develops clear and concise communications to inform stakeholders 2. Utilizes active listening to achieve a shared understanding between communicating parties 3. Distills guidance into simple and actionable language to support mission objectives
The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.		
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	1. Transmit or receive information to enable understanding or redistribution 2. Leverages presentation tools to build and brief ideas 3. Seeks additional information to eliminate misunderstanding or clarify intent
Analytical Thinking Communication Influence Relationship Building		

Figure 4. 1BXXX Communication Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	1. Develops a clear vision to ensure technology enables organizational goals 2. Informs and develops commanders intent to achieve mission objectives 3. Collaborates across organizational boundaries to influence mission outcomes
Sub-Competencies	Advanced Consistency of Application: Sustained application of competency over time in complex situations	1. Fosters a culture of innovation and technical excellence by supporting personnel advancement 2. Translates leadership's intent into actionable objectives to build trust and understanding 3. Leads technical collaborations with stakeholders to support mission objectives 4. Maintains awareness on trends, emerging technologies, and threats to maintain technological relevancy
Leadership		
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	1. Supports, mentors, and empowers personnel to create a professional climate 2. Demonstrates a combination of technical expertise, interpreting strategic vision and possessing leadership skills to ensure mission success 3. Guides subordinates to develop individual technical talent
The ability to guide and influence individuals or groups toward achieving a common goal, encompassing skills like direction, motivation, and decision-making by fostering collaboration, accountability, innovation, technical expertise, and personnel well-being.		
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	1. Maintains individual accountability and encourages peer accountability to successfully execute mission 2. Makes decisions at the lowest level within defined constraints and elevates issues to accelerate mission execution 3. Demonstrates followership by supporting and executing guidance 4. Develops individual technical talent to increase self-competency
Accountability Communication Decision Making Leadership		

Figure 5. 1BXXX Leadership Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Scope: Integration with AF-level/within industry	1. Interprets objectives to posture forces in support of strategic requirements 2. Collaborates with external agencies to influence or shape requirements 3. Creates, updates and implements assessment policies that govern readiness validation to meet mission requirements
Sub-Competencies	Advanced Scope: Integration with organizational strategies	1. Forecasts shortfalls to enable planning and sourcing actions 2. Develops criteria and processes to assess mission capabilities (e.g. people, equipment, KSAs) 3. Leads evaluations and assessments to validate readiness 4. Analyze readiness assessments from multiple sources to deliver readiness breifings
Readiness		
Description	Intermediate Scope: Integration with concerned areas	1. Prepares and tracks personnel and equipment to provide ready forces 2. Provides and advocates for equipment and training to maximize capability readiness 3. Conducts evaluations and assessments to validate capabilities (e.g. people, equipment, KSAs)
The state of preparedness to respond swiftly and effectively to meet mission requirements.		
Supporting Competencies	Basic Scope: Specific Area	1. Maintains individual preparedness and training to meet world-wide requirements (e.g. medical, family, and equipment) 2. Maintains individual mission qualifications to meet requirements 3. Participates in activities to enhance and demonstrate individual proficiency in assigned role (e.g. exercises, training events)
Accountability Flexibility Organizational Awareness Resource Management		

Figure 6. 1BXXX Readiness Competency Rubric

Competency	Proficiency Levels	Observable Behaviors
Operations	Expert Reach of Influence: MAJCOM/AF-Level/Industry	1. Establishes and maintains life-cycle training requirements to ensure career development 2. Advocates and coordinates opportunities to achieve and maintain occupational relevancy 3. Conducts mission analysis to determine training requirements
Sub-Competencies	Advanced Reach of Influence: Wing/Institutional	1. Develops and approves instructional material in accordance with requirements to develop skills, knowledge and abilities 2. Provides foundational and advanced training to meet developmental milestones 3. Assesses and updates training programs to ensure relevance and effectiveness
Training		
Description	Intermediate Reach of Influence: Unit/Groups	1. Reviews and updates training records to document progress 2. Seeks training opportunities to enhance mission execution capabilities 3. Observes and provides feedback to assess trainees' proficiency 4. Trains and develops members in accordance with requirements to enable mission execution
The process of developing Airmen with the knowledge, skills, and abilities necessary for effective mission performance.		
Supporting Competencies	Basic Reach of Influence: Individuals	1. Completes training to meet organizational requirements 2. Applies and utilizes training resources to increase individual skills and abilities 3. Demonstrates basic proficiency for practical application 4. Identifies barriers, issues, and concerns to inform updates
Communication Develops People Information Seeking Initiative		

Figure 7. 1BXXX Training Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Airmanship	Expert Scope: Integration with AF-level/within industry	1. Provides guidance on program implementation to ensure standards compliance 2. Makes decisions and takes actions to ensure successful program implementation 3. Reviews, updates, and verifies policy to align with strategic objectives
Sub-Competencies	Advanced Scope: Integration with organizational strategies	1. Serves as program administrator to ensure compliance 2. Briefs unit command staff on program implementation, compliance and gaps to inform decision making and actions 3. Develops countermeasures to mitigate program risks 4. Tracks assessment outcomes to identify capability gaps (e.g. MICT)
Program Management		
Description	Intermediate Scope: Integration with concerned areas	1. Educates junior members on the applicability of administrative programs to ensure familiarization 2. Serves as alternate administrator to provide program continuity 3. Utilizes digital tools and platforms to improve program efficiency 4. Implements continuous improvement processes to optimize program effectiveness
The management of people through deliberate planning, institutional training, development, direction, and organization to maximize performance.		
Supporting Competencies	Basic Scope: Specific Area	1. Demonstrates familiarization with programs and applicability to ensure individual administrative readiness 2. Follows established procedures, policy, and guidance to meet administrative program requirements 3. Identifies program objectives by reviewing guidance and policy
Accountability Communication Develops People Leadership		

Figure 8. 1BXXX Program Management Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Airmanship	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	1. Formulates strategy to manage manpower 2. Conducts development panels and vectors individuals for key leadership positions 3. Manages requisitions and coordinates change requests to optimize organizational requirements
Sub-Competencies	Advanced Consistency of Application: Sustained application of competency over time in complex situations	1. Establishes policies and manages administrative requirements to enhance effectiveness 2. Identifies personnel constraints and proposes solutions to solve deficiencies 3. Provides feedback and vectors personnel based on performance to inform career advancement (e.g. DT Process)
Personnel Management		
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	1. Documents expectations and feedback to ensure adherence to standards 2. Establishes subordinates responsibilities, impact, and role to support organizational requirements 3. Identifies applicable personnel system resources and available products to inform and manage subordinates 4. Utilizes available tools and programs to highlight individual and team successes
The management of people through deliberate planning, institutional training, development, direction, and organization to maximize performance.		
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	1. Adheres to documented expectations and seeks feedback to ensure compliance to standards 2. Manages tasks to accomplish desired actions or outcomes 3. Articulates responsibilities, impact, and role to demonstrate understanding of position
Accountability Change Management Communication Organizational Awareness		

Figure 9. 1BXXX Personnel Management Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Airmanship	Expert Impact on… AF-level practices/within industry	1. Oversees projects to ensure required resources are available for execution 2. Submits Unfunded Requirements (UFRs), programming inputs, spend/execution plans to support resourcing requirements 3. Demonstrates a comprehensive understanding of Program Objective Memorandum (POM) process, Program Element Codes (PECs), Future Year Defense Program (FYDP) and other funding and resourcing tools to satisfy requirements
Sub-Competencies	Advanced Impact on… Management decisions	1. Evaluates unit projects to determine required resources 2. Secures unit-level resources for project completion 3. Provides impact to unit leadership to mitigate resource shortfalls
Resource Management		
Description	Intermediate Impact on… Specific workplace projects	1. Assesses unit projects for resourcing needs to ensure successful project completion 2. Articulates resourcing shortfalls to ensure successful project completion
The allocation, utilization, and oversight of resources to ensure mission readiness and operational effectiveness.		
Supporting Competencies	Basic Impact on… Specific workplace tasks	1. Identifies resources required to perform basic unit tasks 2. Identifies resourcing shortfalls to inform leadership situational awareness
Accountability Decision Making Organizational Awareness Resource Management		

Figure 10. 1BXXX Resource Management Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Mission Execution	Expert Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area	1. Identifies, creates, or integrates emerging capabilities to achieve overmatch within the environment 2. Provides technical expertise to advise stakeholders
Sub-Competencies	Advanced Depth of Knowledge: New practices of all workplace elements	1. Develops TTPs based on environmental conditions and actors to achieve mission objectives 2. Determines detection or subversion methods to impact mission-critical signals 3. Modifies existing equipment functions, adapting to dynamic mission requirements
Electromagnetic Spectrum		
Description	Intermediate Depth of Knowledge: Established practices of all workplace elements	1. Use and explain available frequencies within the environment to gain tactical advantage 2. Extracts, generates, or inhibits signals appropriately for effective execution of tactical plans 3. Selects, configures, and troubleshoots equipment needed to interact with the EMS
The aggregate of knowledge and application of technologies employing electromagnetic radiation, signals and transmissions, tradecraft, and production of effects.		
Supporting Competencies	Basic Depth of Knowledge: Established practice with some workplace elements	1. Characterizes different frequencies on the EMS to determine application 2. Determines the frequencies, channels, transmission rates, and bandwidth of signals of interest to characterize activity 3. Identifies physical equipment utilized to interact with the EMS 4. Explains the interaction of emissions within the environment to support mission execution
Analytical Thinking Creative Thinking Digital Literacy Information Seeking		

Figure 11. 1BXXX Electromagnetic Spectrum Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Mission Execution	Expert Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area	1. Creates or integrates emerging firmware capabilities to develop solutions 2. Envisions, engineers, and fabricates purpose-specific hardware solutions to maintain competitive advantage 3. Provides technical expertise to advise stakeholders
Sub-Competencies Hardware	Advanced Depth of Knowledge: New practices of all workplace elements	1. Identifies, obtains, and integrates appropriate hardware solutions to meet mission needs 2. Leverages or incorporates available external infrastructure to enhance mission effectiveness 3. Seeks, architects, and implements optimizations to redistribute hardware workloads across mission infrastructure 4. Enhances, repurposes, or changes firmware functionality to meet mission needs
Description The aggregate of knowledge and application of technologies employing tangible components and firmware, defining infrastructure across physical and virtualized levels.	Intermediate Depth of Knowledge: Established practices of all workplace elements	1. Deploys or swaps hardware, interfaces, or components to enable data paths, change capacity, or enable capability 2. Converts or scales resources between computing environments (e.g. physical, virtual, or containerized) to meet mission requirements 3. Articulates capability requirements and limitations to support decision making 4. Applies structured diagnostic techniques to troubleshoot common hardware and firmware issues
Supporting Competencies Analytical Thinking Creative Thinking Digital Literacy Information Seeking	Basic Depth of Knowledge: Established practice with some workplace elements	1. Describes hardware components to articulate functional and mission purpose 2. Follows specified hardware maintenance procedures to ensure operational function 3. Identifies hardware states to determine troubleshooting requirements 4. Explains components of infrastructures, physical and virtual, for different use cases

Figure 12. 1BXXX Hardware Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Mission Execution	Expert Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area	1. Provides technical expertise to advise stakeholders 2. Creates or integrates emerging capabilities to achieve overmatch within the network 3. Collect, characterizes, and affects proprietary or specialized network data to support mission objectives 4. Envisions, advocates for, or engineers network architecture to optimize or achieve overmatch
Sub-Competencies Networking	Advanced Depth of Knowledge: New practices of all workplace elements	1. Applies techniques to resolve complex networking issues 2. Uses network capabilities and access methods to automate, optimize, or obfuscate interactions 3. Develops TTPs adapting to dynamic mission requirements 4. Modifies, adapts, or integrates existing capabilities to accomplish mission objectives
Description The aggregate of knowledge and technologies employing the logical interconnection of devices to enable information transfer between systems.	Intermediate Depth of Knowledge: Established practices of all workplace elements	1. Configures and troubleshoots existing network capabilities to modify functionality 2. Articulates and assesses network activity to support decision making 3. Selects and determines appropriate locations to employ network capabilities
Supporting Competencies Analytical Thinking Creative Thinking Digital Literacy Information Seeking	Basic Depth of Knowledge: Established practice with some workplace elements	1. Examines network traffic to characterize activity 2. Selects and uses network traffic capabilities to meet mission needs 3. Develops and interprets network maps to define the environment

Figure 13. 1BXXX Networking Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Mission Execution	Expert Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area	1. Evaluates, creates, integrates emerging or novel concepts and TTPs to improve mission efficiency, accuracy, and adaptability 2. Creates, innovates, or modifies software and platforms beyond the engineered intent 3. Provides technical expertise to advise stakeholders
Sub-Competencies	Advanced Depth of Knowledge: New practices of all workplace elements	1. Develops or reverse-engineers software to support the mission 2. Deploys system capabilities and access methods to automate, optimize, or obfuscate interactions 3. Seeks, architects, and implements optimizations to adapt software for complex scenarios 4. Develops TTPs adapting to dynamic mission requirements
Software		
Description	Intermediate Depth of Knowledge: Established practices of all workplace elements	1. Analyzes and configures a script or program to mitigate vulnerabilities or modify functionality 2. Articulates and assesses system activity to support decision making 3. Selects and determines appropriate locations to employ system or software capabilities 4. Deploys, configures, and troubleshoots solutions with existing capabilities to control operating environment
The aggregate of knowledge and technologies to develop the logical instruction sets, data, or programs that tell a system what to do and how to perform specific tasks.		
Supporting Competencies	Basic Depth of Knowledge: Established practice with some workplace elements	1. Articulates software structures to characterize functionality 2. Builds or troubleshoots scripts and programs to complete tasks 3. Identifies concepts (e.g. code and security) across various languages, computer architectures, and platforms to define the environment 4. Explains the orchestration of system processes alongside component integration to achieve mission requirements
Analytical Thinking Creative Thinking Digital Literacy Information Seeking		

Figure 14. 1BXXX Software Competency Rubric.

Competency	Proficiency Levels	Observable Behaviors
Mission Execution	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	1. Evaluates, creates, integrates emerging or novel concepts to improve mission efficiency, accuracy, and adaptability 2. Formulates and establishes organizational analytical techniques to improve mission outcomes 3. Envisions, advocates for, or engineers data architecture (sensors, links, storage, processing) to optimize or achieve overmatch
Sub-Competencies	Advanced Consistency of Application: Sustained application of competency over time in complex situations	1. Develops and refines workflows to improve decision making 2. Applies or modifies sophisticated analytical methods to address mission scenarios 3. Identifies, obtains, and integrates new data sources to refine or improve decision-making 4. Optimizes or integrates existing data architectures (e.g. sensors, links, storage, processing) to improve mission outcomes
Analysis		
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	1. Depicts activities in the operational environment using visualizations to support further analysis or decision-making 2. Conducts data actions (e.g. clean, transform, and fuse) on existing data sources for enrichment, storage, and integration into information flows 3. Correlates mission objectives with analytical conclusions to identify risks and intelligence gaps 4. Documents and reports findings from analysis to support decision-making
The use of methods and technologies to conduct data analysis (collect, clean, compile, visualize, present, evaluate) to produce information.		
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	1. Establish data flows to process information in support of analysis 2. Queries and scans systems to return mission-related information or data 3. Applies analytical methods on data and information to draw conclusions 4. Describes analytical conclusions to stakeholders
Analytical Thinking Data Interpretation Digital Literacy Information Seeking		

Figure 15. 1BXXX Analysis Competency Rubric.

12. Specialty Qualification Requirements.

12.1. Apprentice (3-Level) Training.

KNOWLEDGE	Computer Operating Systems, Software Applications, Database Concepts, Common Programming Languages, Hardware Components, Networking Fundamentals, Protocols, Network Addressing, Network Infrastructure, Telecommunications Theory, Data Communications, Wireless Technologies, Cryptography and Cyber Operation Laws
EDUCATION	For entry into this specialty, completion of high school is mandatory. Additional courses in Science, Technology, Engineering, and Mathematics (STEM) is desirable. Associate degree or higher in related fields or Information Technology (IT) Certification is desirable.
TRAINING	Completion of the Cyber Warfare Operations Apprentice course (See Part II, Section B for Course Objective List)
EXPERIENCE	None required
OTHER	For entry into this specialty: Must earn a minimum score of 68 on the two-factor model that includes the Armed Services Vocational Aptitude Battery (ASVAB) and Tailored Adaptive Personality Assessment System (TAPAS) scores.Requires routine access to Top Secret material or similar environment, completion of a current Single Scope Background Investigation (SSBI) according to DoDMAN5200.02_DAFMAN16-1405, *Department of the Air Force Personnel Security Program*, is mandatory for award and retention of this skill level. **NOTE:** Award of the 3-skill level without a completed SSBI is authorized provided an interim Top Secret clearance has been granted according to DoDMAN5200.02_DAFMAN16-1405. For award and retention of 1B431, individual must maintain local network access IAW AFI 17-130, *Cybersecurity Program Management* and DAFMAN 17-1301, *Computer Security (COMPUSEC).*
IMPLEMENTATION	Attendance at the Cyber Warfare Operations Apprentice Course is mandatory for award of the 3-skill level unless waived by the 1B AFCFM

12.2. Journeyman (5-Level) Training.

KNOWLEDGE	All 1B431 knowledge qualifications apply to the 1B451 requirements
TRAINING	Completion of the 1B451 Career Development Course if applicable. Completion of all STS core tasks. Completion of applicable AFJQS/AFQTPs. Completion of all local tasks assigned for the duty position to include Crew Position Certification if required for duty position.
EXPERIENCE	Qualification in and possession of AFSC 1B431 Experience performing Cyber Warfare Operations functions as outlined in Section B, para. 4.
OTHER	Requires routine access to Top Secret material or similar environment, completion of a current Single Scope Background Investigation (SSBI) according to AFI 31-501, *Personnel Security Program Management*, is mandatory for award and retention of this skill level. For award and retention of 1B451, individual must maintain local network access IAW AFI 17-130, *Cybersecurity Program Management* and AFMAN 17-1301, *Computer Security (COMPUSEC).*
IMPLEMENTATION	Entry into formal journeyman upgrade training is accomplished once individuals are assigned to their first duty station. Qualification training is initiated anytime individuals are assigned duties for which they are not qualified. Use OJT, CDCs, CFETP, CBTs, and AFJQSs/AFQTPs concurrently to obtain the necessary qualification for refresher and cross-utilization training.

12.3. Craftsman (7-Level) Training.

KNOWLEDGE	All 1B451 knowledge qualifications apply to the 1B471 requirements
TRAINING	Completion of the 1B471 Career Development Course if applicable. Completion of all STS core tasks. Completion of applicable AFJQS/AFQTPs. Completion of all local tasks assigned for the duty position to include Crew Position Certification if required for duty position.
EXPERIENCE	Qualification in and possession of AFSC 1B451 Experience performing or supervising Cyber Warfare Operations functions as outlined in Section B, para. 4.
OTHER	Requires routine access to Top Secret material or similar environment, completion of a current Single Scope Background Investigation (SSBI) according to AFI 31-501, *Personnel Security Program Management*,is mandatory for award and retention of this skill level. For award and retention of 1B471, individual must maintain local network access IAW AFI 17-130, *Cybersecurity Program Management* and DAFMAN 17-1301, *Computer Security (COMPUSEC).*
IMPLEMENTATION	Entry into OJT is initiated when individuals obtain the necessary rank and skill level. Qualification training is initiated anytime an individual is assigned duties for which they are not qualified. Use OJT, CBTs, CDCs, CFETP, and AFJQSs/AFQTPs concurrently to obtain the necessary qualification for refresher and cross-utilization training.

12.4. Superintendent (9-Level) Training

KNOWLEDGE	Resource Management, Manpower and Organization, Training Management, Deployment Management, and Base/Unit Functional Management
TRAINING	None
EXPERIENCE	Qualification in and possession of AFSC 1B471 Managing and directing Cyber Warfare Operations personnel and processes.
OTHER	Requires routine access to Top Secret material or similar environment, completion of a current Single Scope Background Investigation (SSBI) according to AFI 31-501, *Personnel Security Program Management*, is mandatory for award and retention of this skill level.
IMPLEMENTATION	Entry into OJT is initiated when individuals are selected for the rank of SMSgt. Qualification training is initiated anytime individuals are assigned duties for which they are not qualified.

12.5. Training Sources.

12.5.1. AFSC Specific Training – 333 TRS, Keesler AFB, MS at https://usaf.dps.mil/teams/app10-etca/SitePages/Home.aspx, course ID: E2ABR1B431 0A1A

12.5.2. 1B4X1 CDCs if available, will be available through the Unit Training Manager. Once the individual has been validated and enrolled into the program, he or she will receive a link to the online CDCs. Paper-based CDCs are no longer supported.

12.5.3. AFJQSs/AFQTPs are Air Force publications and are mandatory for use by personnel in upgrade or qualification training. They are developed by the 81 TRSS (Q-Flight), Keesler AFB, MS and may be downloaded from https://usaf.dps.mil/teams/10445

12.5.3.1. Procedures for requesting development of AFJQSs/AFQTPs are contained in AFMAN 17-204, *Air Force On-the-Job Training Products for Cyberspace Support Enlisted Specialty Training*. AFJQSs/AFQTPs are listed in Part II, Section C, of this CFETP.

Section D - Resource Constraints

13. Purpose. This section identifies known resource constraints that preclude optimal/desired training from being developed or conducted, including information such as cost and manpower. Included are narrative explanations of each resource constraint and an impact statement describing what effect each constraint has on training, the resources needed, and actions required to satisfy the training requirements.

14. Apprentice (3-Level) Training.

14.1. Impact. Constrained course throughput capacity.

14.2. Resources Required. Additional Instructors.

14.3. Action Required. Coordination with Air Reserve Component (ARC) and Space Force for additional cadre.

14.4. OPR/Target Completion Date. 333 TRS/FY27

15. Journeyman (5-Level) Training.

15.1. Impact. None.

15.2. Resources Required. None.

15.3. Action Required. None.

15.4. OPR/Target Completion Date. None.

16. Craftsman (7-Level) Training.

16.1. Impact. None.

16.2. Resource Required. None.

16.3. Action Required. None.

16.4. OPR/Target Completion Date.

17. Superintendent (9-Level) Training.

17.1. Impact. None.

17.2. Resource Required. None.

17.3. Action Required. None.

17.4. OPR/Target Completion Date. None.

Section E - Transition Training Guide

There are currently no transition training requirements. This area is reserved.

PART II

Section A - Specialty Training Standard

1. Implementation. The STS (attached) will be used for technical training provided by AETC

2. Purpose.

2.1. Column A (Competencies, Behaviors, and Learning Outcomes) lists the competencies, required behaviors and learning outcomes associated with the 1BXXX Occupational Competency Model to perform at the Basic, Intermediate, Advanced, and Expert proficiency levels.

2.2. Column B (Behavior Match) indicates which of the required behaviors listed in the competency heading the learning outcome is associated with.

2.3. Column C (Core/Cert) identifies learning outcomes requiring third party certification with a "C".

2.4. NOTE: (Wartime) This column isn't used for the 1BXXX AFSC, all identified 3-level requirements are Wartime course requirements.

2.5. Column D (Deployment */SEI +/CBRN ♦) Learning Outcomes identified with an (*) are competencies Airmen should be qualified on (per their skill level) prior to deployment. Items identified with a (+) are reserved for the award of a specific SEI. There are currently no outcomes identified by the CFM with an (*) or (+).

2.5.1. Learning outcomes identified with an (♦) require annual CBRN (Training Task Qualification) training in the work center. Per DAFI 10-2503, para 6.5.1.1., CBRN Defense TQT is defined as a hands-on event in MOPP 4 gear performing regular duties. Unit Commanders will identify tasks within their Master Training Plans (MTP) and ensure Airmen can perform these tasks if chemical, biological, radiological or nuclear (CBRN) warfare hazards are present based on their mission requirements.

2.6. Columns J-M (3-Lvl, 5-Lvl, 7-Lvl, 9-Lvl) identify learning outcome requirements by skill-level. A behavioral proficiency code of "K", "P", or "kp" in the column indicates the outcome is taught in a formal course to that proficiency.

Behavioral Statement STS Coding System	
Code	**Definition**
K	Subject Knowledge Training - The verb selection identifies the individual's ability to identify facts, state principles, analyze or evaluate the subject.
P	Performance Training - Identifies that the individual has performed the task to the satisfaction of the course; however, the individual may not be capable of meeting the filed requirements for speed and accuracy.
pk	Performance Knowledge Training - The verb selection identifies the individual's ability to relate simple facts, procedures, operating principles and operational theory for the task.
-	No training provided in the formal course or DL course.
X	Training is required but not provided due to limitations in resources.

2.7. Becomes a job qualification standard (JQS) for on-the-job training when placed in AF Form 623, Individual Training Record folder, and used according to DAFMAN 36-2689.

3. Recommendations. Comments and recommendations are invited concerning the quality of AETC training. A Training Feedback Hotline has been installed for the supervisors' convenience. For a quick response to concerns, call our Training Feedback Hotline at DSN 597-4566, or e-mail us at 81TRG.TGE.Workflow@us.af.mil. Reference this STS and identify the specific area of concern (paragraph, training standard element, etc).

BY ORDER OF THE SECRETARY OF THE AIR FORCE

OFFICIAL

MICHELLE N. LORDEN, CMSgt, USAF
1BXXX Career Field Manager

Attachments:
1. 1B4X1 STS

THIS BLOCK IS FOR IDENTIFICATION PURPOSES ONLY Personal Data – Privacy Act of 1974		
PRINTED NAME OF TRAINEE *(Last, First, Middle Initial)*	INITIALS *(Written)*	LAST 4 OF SSAN
PRINTED NAME OF TRAINER AND CERTIFYING OFFICIAL AND WRITTEN INITIALS		
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	
N/I	N/I	

Task, Knowledge, and Proficiency Level

Code	Definition
K	Subject Knowledge Training - The verb selection identifies the individual's ability to identify facts, state principles, analyze, or evaluate the subject.
P	Performance Training - Identifies that the individual has performed the task to the satisfaction of the course; however, the individual may not be capable of meeting the field requirements for speed and accuracy.
pk	Performance Knowledge Training - The verb selection identifies the individual's ability to relate simple facts, procedures, operating principles, and operational theory for the task.
-	No training provided in the course or CDC.
X	Training is required but not provided due to limitations in resources.
Each STS element is written as a behavioral statement. The detail of the statement and verb selection reflects the level of training provided by resident training and CDCs.	

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1. Operations												
1.1 PLANNING **Description: The practice of translating guidance into actionable tasks by balancing stakeholder needs and risks in order to achieve mission objectives.** **Supporting Competencies:** *Analytical Thinking, Communication, Decision Making, Strategic Thinking* **TR:** AFI 16-1007, AFTTP 3-10.3, CJCSI 5123.01, CJCSM 3130.02, CJCSM 3500.03, DAFI 17-201, DAFPD 17-2, JP 1-0, JP 2-0, JP 3-12, JP 3-85, JP 3-13.3, JP 3-52 Ch 3-d, JP 3-60, JP 5-0											
1.1.1. PLANNING (BASIC) **Required Behaviors:** 1. Identifies authoritative sources and documents to inform planning 2. Interprets authoritative sources and documents to identify mission parameters 3. Applies appropriate planning methods to achieve objectives 4. Crafts planning products to guide execution Criteria: Consistency of Application: Sustained application of competency over time												
1.1.1.1. Identify order types	1	*							**K**	-	-	-
1.1.1.2. Identify strategic guidance	1	*							**K**	-	-	-
1.1.1.3. Align products to their planning purposes	1	*							**K**	-	-	-
1.1.1.4 Identify Intelligence Programs	1	*							**K**	-	-	-
1.1.1.5. Derive mission parameters from authoritative sources	2	*							**K**	-	-	-
1.1.1.6. Articulate tasks from authoritative source	2								-	-	-	-
1.1.1.7. Identify key stakeholders	2	*							**K**	-	-	-
1.1.1.8. Describe DoW infrastructure and defensive strategies	2	*							**K**	-	-	-
1.1.1.9. Identify SIGINT systems, architectures, and associated reporting/tracking processes	2	*							**K**	-	-	-
1.1.1.10. Identify application of planning methodologies	3	*							**K**	-	-	-
1.1.1.11. Apply steps of planning methodologies	3	*							**P**	-	-	-
1.1.1.12. Develop planning deliverables	4	*							**P**	-	-	-
1.1.1.13. Present products for review and approval	4	*							**P**	-	-	-
1.1.2. PLANNING (INTERMEDIATE) **Required Behaviors:** 1. Leads small teams through planning cycle to meet planning objectives 2. Coordinates with mission partners to integrate effects 3. Refines planning to adapt to the operational environment 4. Assess effectiveness post execution to inform planning Criteria: Scope: Consistency of Application: Sustained application of competency over time in a variety of situations												
1.1.2.1. Direct team actions	1								-	-	-	-
1.1.2.2. Coordinate team products and outcomes	1								-	-	-	-
1.1.2.3. Identify capabilities that support mission objectives	2	*							**K**	-	-	-
1.1.2.4. Integrate partner effects into synchronization tools (e.g., fires plan, CONOP)	2								-	-	-	-
1.1.2.5. Assess changes in the operational environment that impact planning decisions	3	*							**pk**	-	-	-
1.1.2.6. Revise mission tasks or phases in response to changing operational conditions	3								-	-	-	-
1.1.2.7. Conduct post mission analysis	4	*							**P**	-	-	-
1.1.2.8. Document lessons learned	4	*							**P**	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.1.3. PLANNING (ADVANCED) **Required Behaviors:** 1. Leverages alternate planning models for complex planning 2. Leads integrated planning to develop viable operational approaches 3. Leverages available authorities to achieve desired effects Criteria: Consistency of Application: Sustained application of competency over time in complex situations												
1.1.3.1. Identify complementary planning methodologies	1								-	-	-	-
1.1.3.2. Leverage complementary planning methodologies	1								-	-	-	-
1.1.3.3. Develop approaches leveraging operational art/design	2								-	-	-	-
1.1.3.4. Integrate Joint, Inter-Agency, Inter-Governmental, Multi-National (JIIM) capabilities	2								-	-	-	-
1.1.3.5. Perform deconfliction actions	2								-	-	-	-
1.1.3.6. Identify authorities needed	3	*							**K**	-	-	-
1.1.3.7. Request authorities	3								-	-	-	-
1.1.3.8. Integrate mission-partner authorities	3								-	-	-	-
1.1.4. PLANNING (EXPERT) **Required Behaviors:** **1. Designs training and exercises to increase operational effectiveness** **2. Guides authoritative documents and sources used for planning** **3. Guides requirements process to meet operational needs** **Criteria: Scope: Consistency of Application:** **Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency**												
1.1.4.1. Develop learning objectives	1								-	-	-	-
1.1.4.2. Integrate capabilities, threats, technologies, and/or operational lessons into exercise design	1								-	-	-	-
1.1.4.3. Identify shortfalls in planning documents and sources	2								-	-	-	-
1.1.4.4. Author planning guidance	2								-	-	-	-
1.1.4.5. Comprehend the requirements process	3								-	-	-	-
1.1.4.6. Perform capability gap analysis	3								-	-	-	-
1.1.4.7. Formulate requirements	3								-	-	-	-
1.2. COMMUNICATION **Description: The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.** **Supporting Competencies:** *Analytical, Communication, Influence, Relationship Building* **TR:** AFH 1-1 Ch 15, AFH 33-337; AFMAN 33-326, CJCSM 6510.01F, JP 3-12, JP 5-0, JP 6-0												
1.2.1. COMMUNICATION (BASIC) **Required Behaviors:** **1. Transmit or receive information to enable understanding or redistribution** **2. Leverages presentation tools to build and brief ideas** **3. Seeks additional information to eliminate misunderstanding or clarify intent** **Criteria: *Consistency of Application: Sustained application of competency over time***												
1.2.1.1. Interpret information received from an email, slide brief, and verbal brief	1	*							**K**	-	-	-
1.2.1.2. Deliver communications utilizing email, slide brief, and verbal brief	1	*							**P**	-	-	-
1.2.1.3. Summarize key information from relevant mission documents (e.g., OPORD, RFI, Incident report, AAR, etc.)	1	*							**K**	-	-	-
1.2.1.4. Select appropriate COMREL communications and feedback channel	1	*							**pk**	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.2.1.5. List examples of traditional and emerging communication tools	2	*							K	-	-	-
1.2.1.6. Utilize a presentation tool to build and conduct briefing	2	*							P	-	-	-
1.2.1.7. Solicit feedback	3	*							-	P	-	-
1.2.1.8. Resolve communication gaps through feedback	3	*							-	P	-	-
1.2.2. COMMUNICATION (INTERMEDIATE) **Required Behaviors:** **1. Develops clear and concise communications to inform stakeholders** **2. Utilizes active listening to achieve a shared understanding between communicating parties** **3. Distills guidance into simple and actionable language to support mission objectives** **Criteria:** ***Consistency of Application: Sustained application of competency over time in a variety of situations***												
1.2.2.1. Analyze communications for clarity and brevity	1	*							K	-	-	-
1.2.2.2. Revise communications for clarity and brevity	1	*							P	-	-	-
1.2.2.3. Develop stakeholder briefs	1	*							P	-	-	-
1.2.2.4. Deliver stakeholder briefs	1	*							P	-	-	-
1.2.2.5. Utilize appropriate communications platforms for stakeholders	1								-	-	-	-
1.2.2.6. Recall information received from active listening	2	*							P	-	-	-
1.2.2.7. Construct new communication content based on received information	2	*							P	-	-	-
1.2.2.8. Extract relevant components from guidance	3	*							P	-	-	-
1.2.2.9. Compile extracted components into actionable tasks	3	*							P	-	-	-
1.2.3. COMMUNICATION (ADVANCED) **Required Behaviors:** **1. Compiles information from various sources to shape operational activities** **2. Simplifies technical information into clear and concise language to ensure comprehension** **3. Distills guidance into actionable language to support mission objectives** **Criteria:** ***Consistency of Application: Sustained application of competency over time in complex situations***												
1.2.3.1. Gather information for operational use	1	*							-	P	-	-
1.2.3.2. Identify information sources for operational use	1	*							-	K	-	-
1.2.3.3. Fuse data from multiple information sources	1	*							-	P	-	-
1.2.3.4. Examine products and provide feedback	1	*							-	P	-	-
1.2.3.5. Translate complex topics to simplified tasks for subordinates	2	*							-	-	pk	-
1.2.3.6. Translate technical information into business language for senior decision makers	2	*							-	-	P	-
1.2.3.7. Explain impacts of technical advantages and deficiencies to non-technical leaders	2	*							-	-	P	-
1.2.3.8. Evaluate mission guidance	3	*							-	K	-	-
1.2.3.9. Employ organization processes and products to effect or recommend change	3								-	-	-	-
1.2.3.10. Consolidate stakeholder inputs and guidance into actionable tasks	3								-	-	-	-
1.2.4. COMMUNICATION (EXPERT) **Required Behaviors:** **1. Authors and employs strategies (e.g. CONOPS) to address the unique needs of stakeholders** **2. Conveys information in actionable language to support mission objectives** **3. Facilitates collaboration between stakeholders to integrate and achieve outcomes** **Criteria:** ***Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency***												
1.2.4.1. Solicit stakeholder needs & deficiencies	1								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.2.4.2. Compile stakeholder needs & deficiencies	1								-	-	-	-
1.2.4.3. Shape the development of tactical, operational, or strategic guidance	1								-	-	-	-
1.2.4.4. Formulate guidance	2								-	-	-	-
1.2.4.5. Adapt communications data into mission essential information	2								-	-	-	-
1.2.4.6. Lead working groups with relevant stakeholders	3								-	-	-	-
1.2.4.7. Formulate actionable outcomes from stakeholder objectives	3								-	-	-	-
1.3. LEADERSHIP **Description: The appropriate presentation, promotion, and conveyance of ideas through active listening, messaging, and translation of relevant information.** **Supporting Competencies:** *Accountability, Communication, Decision Making, Leadership* **TR:** AFH 1-1 Ch 14, DAFI 90-1101, DAFI 90-1601, DAFMAN 36-2100, JP 5-0 Ch 5, The Enlisted Force Structure												
1.3.1. LEADERSHIP (BASIC) **Required Behaviors:** **1. Maintains individual accountability and encourages peer accountability to successfully execute mission** **2. Makes decisions at the lowest level within defined constraints and elevates issues to accelerate mission execution** **3. Demonstrates followership by supporting and executing guidance** **4. Develops individual technical talent to increase self-competency** **Criteria:** ***Consistency of Application: Sustained application of competency over time***												
1.3.1.1. Identify personal responsibilities	1	*							K	-	-	-
1.3.1.2. Identify team requirement	1	*							K	-	-	-
1.3.1.3. Explain negative impacts of not being accountable	1	*							K	-	-	-
1.3.1.4. Examine mission risks	2	*							-	pk	-	-
1.3.1.5. Select acceptable risk within scope of responsibility	2	*							-	P	-	-
1.3.1.6. Describe process for elevating risk to supervision	2	*							K	-	-	-
1.3.1.7. Make decisions within defined limits	2	*							P	-	-	-
1.3.1.8 Apply Methodologies of decision making	2	*							K			
1.3.1.9. Follow directions	3	*							P	-	-	-
1.3.1.10. Use guidance for mission accomplishment	3	*							P	-	-	-
1.3.1.11. Identify knowledge and performance gaps	4	*							-	K	-	-
1.3.1.12. Assess training plan	4	*							-	K	-	-
1.3.1.13. Summarize avenues for additional training opportunities	4	*							-	K	-	-
1.3.2. LEADERSHIP (INTERMEDIATE) **Required Behaviors:** **1. Supports, mentors, and empowers personnel to create a professional climate** **2. Demonstrates a combination of technical expertise, interpreting strategic vision and possessing leadership skills to ensure mission success** **3. Guides subordinates to develop individual technical talent** **Criteria:** ***Consistency of Application: Sustained application of competency over time in a variety of situations***												
1.3.2.1. Promote peer accountability	1	*							K	-	-	-
1.3.2.2. Initiate timely feedback	1	*							-	P	-	-
1.3.2.3. Ensure mission objectives align with strategic vision	2	*							-	K	-	-
1.3.2.4. Develop metrics to measure goal accomplishment	2	*							-	P	-	-
1.3.2.5. Lead mission planning, briefing, execution and debriefing	2	*							-	-	pk	-
1.3.2.6. Apply metrics and guidance during mission execution	2	*							-	P	-	-
1.3.2.7. Examine subordinate's technical aptitude during task performance	3	*							-	-	pk	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.3.2.8. Relate feedback to mission objectives	3	*							-	-	pk	-
1.3.2.9. Explain developmental expectations to subordinates	3								-	-	-	-
1.3.2.10. Develop training plans for subordinates	3	*							-	-	pk	-
1.3.3. LEADERSHIP (ADVANCED) **Required Behaviors:** **1. Fosters a culture of innovation and technical excellence by supporting personnel advancement** **2. Translates leadership's intent into actionable objectives to build trust and understanding** **3. Leads technical collaborations with stakeholders to support mission objectives** **4. Maintains awareness on trends, emerging technologies, and threats to maintain technological relevancy** **Criteria:** ***Consistency of Application: Sustained application of competency over time in complex situations***												
1.3.3.1. Implement development strategies that cultivate innovation and technical growth across multi-functional teams	1								-	-	-	-
1.3.3.2. Lead initiatives that enhance personnel expertise and promote a culture of continuous improvement	1								-	-	-	-
1.3.3..3 Develop lines of effort from mission objectives	2								-	-	-	-
1.3.3.4. Explain guidance to subordinates	2	*							-	-	K	-
1.3.3.5. Explain importance of accomplishing objectives	2	*							-	-	K	-
1.3.3.6. Establish relationships with appropriate stakeholders	3								-	-	-	-
1.3.3.7. Identify governance aligned to mission	3								-	-	-	-
1.3.3.8. Facilitate working groups with stakeholders	3								-	-	-	-
1.3.3.9. Identify mission and capability gaps	4	*							-	-	-	K
1.3.3.10. Compare advantages and disadvantages of emerging technologies	4								-	-	-	-
1.3.3.11. Incorporate relevant best practices and lessons learned from outside organizations	4								-	-	-	-
1.3.4. LEADERSHIP (EXPERT) **Required Behaviors:** **1. Develops a clear vision to ensure technology enables organizational goals** **2. Informs and develops commanders intent to achieve mission objectives** **3. Collaborates across organizational boundaries to influence mission outcomes** **Criteria:** ***Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency***												
1.3.4.1. Formulate long-term technology integration strategies	1								-	-	-	-
1.3.4.2. Guide others in developing mission-focused technology roadmaps	1								-	-	-	-
1.3.4.3. Translate guidance into organizational vision	1								-	-	-	-
1.3.4.4. Drive future capability development	1								-	-	-	-
1.3.4.5. Translate commander's intent into actions	2	*							-	-	-	P
1.3.4.6. Formulate intended end state	2								-	-	-	-
1.3.4.7. Integrate modernization strategies into the organization	2								-	-	-	-
1.3.4.8. Build partnerships across organizations	3								-	-	-	-
1.3.4.9. Drive inter-organizational change or consensus	3								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.4. READINESS **Description: The state of preparedness to respond swiftly and effectively to meet mission requirements.** **Supporting Competencies:** ***Accountability, Flexibility, Organizational Awareness, Resource Management*** **TR:** AFFORGEN Connect (https://envision.af.mil/afc), AFI 10-401, AFI 10-403, CJCSM 3130.06, CJCSM 3150.02, DAFI 10-201, DAFI 36-2670, DAFMAN 36-2689, JP 3-08, JP 4-0												
1.4.1. READINESS (BASIC) **Required Behaviors:** **1. Maintains individual preparedness and training to meet world-wide requirements (e.g. medical, family, and equipment)** **2. Maintains individual mission qualifications to meet requirements** **3. Participates in activities to enhance and demonstrate individual proficiency in assigned role (e.g. exercises, training events)** **Criteria: Scope: Integration with concerned areas**												
1.4.1.1. Identify mobility requirements	1	*							-	**K**	-	-
1.4.1.2. Articulate the potential consequences of failing to maintain individual mobility readiness	1	*							-	**K**	-	-
1.4.1.3. Identify the mission-specific qualifications and certifications required within assigned role	2	*							-	**K**	-	-
1.4.1.4. Describe the procedures and timelines for completing, renewing, and tracking mission qualification requirements	2	*							-	**K**	-	-
1.4.1.5. Differentiate between qualification and certification	2	*							-	**K**	-	-
1.4.1.6. Identify applicable training and exercises	3	*							-	**K**	-	-
1.4.1.7. Participate in applicable training and exercises	3								-	-	-	-
1.4.1.8. Explain the purpose and importance of participating in proficiency-building activities	3								-	-	-	-
1.4.2. READINESS (INTERMEDIATE) **Required Behaviors:** **1. Prepares and tracks personnel and equipment to provide ready forces** **2. Provides and advocates for equipment and training to maximize capability readiness** **3. Conducts evaluations and assessments to validate capabilities (e.g. people, equipment, KSAs)** **Criteria:** ***Scope: Integration with concerned areas***												
1.4.2.1. Utilize established readiness tracking systems (e.g., JCC2-R, DRRS, etc.)	1								-	-	-	-
1.4.2.2. Shape requirements for equipment and training	1								-	-	-	-
1.4.2.3. Monitor personnel and equipment status in support of force presentation	1								-	-	-	-
1.4.2.4. Enforce personnel readiness training requirements	1	*							-	-	**P**	-
1.4.2.5. Provide assigned-role training to personnel	2								-	-	-	-
1.4.2.6. Advocate for additional or upgraded equipment and training resources based on mission requirements	2								-	-	-	-
1.4.2.7. Validate compliance	3								-	-	-	-
1.4.2.8. Recommend corrective actions based on trends, gaps, or recurring deficiencies	3								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.4.3. READINESS (ADVANCED) **Required Behaviors:** **1. Forecasts shortfalls to enable planning and sourcing actions** **2. Develops criteria and processes to assess mission capabilities (e.g. people, equipment, KSAs)** **3. Leads evaluations and assessments to validate readiness** **4. Analyze readiness assessments from multiple sources to deliver readiness briefings** **Criteria: Scope: Integration with organizational strategies**												
1.4.3.1. Analyze historical data, current readiness trends, and projected mission demands	1								-	-	-	-
1.4.3.2. Evaluate mitigation plans	1								-	-	-	-
1.4.3.3. Design assessment frameworks and metrics	2								-	-	-	-
1.4.3.4. Refine capability assessment processes	2								-	-	-	-
1.4.3.5. Conduct readiness assessments	3								-	-	-	-
1.4.3.6. Implement changes that improve readiness postures	3								-	-	-	-
1.4.3.7. Consolidate readiness data from available reporting systems (e.g., DRRS, SORTS, JCC2-R, local trackers, etc.)	4								-	-	-	-
1.4.3.8. Translate complex readiness data into actionable insights	4								-	-	-	-
1.4.4. READINESS (EXPERT) **Required Behaviors:** **1. Interprets objectives to posture forces in support of strategic requirements** **2. Collaborates with external agencies to influence or shape requirements** **3. Creates, updates and implements assessment policies that govern readiness validation to meet mission requirements** **Criteria:** ***Scope: Integration with AF-level/within industry***												
1.4.4.1. Interpret force-wide forecasting models	1								-	-	-	-
1.4.4.2. Project readiness gaps	1								-	-	-	-
1.4.4.3. Advise stakeholders on forecasted shortfalls	1								-	-	-	-
1.4.4.4. Coordinate interagency requirements	2								-	-	-	-
1.4.4.5. Develop partnerships with interagency, multinational, and industry stakeholders	2								-	-	-	-
1.4.4.6. Shape readiness policy	3								-	-	-	-
1.4.4.7. Institutionalize capability assessment frameworks	3								-	-	-	-
1.5. TRAINING **The process of developing Airmen with the knowledge, skills, and abilities necessary for effective mission performance.** **Supporting Competencies:** ***Communication, Develops People, Information Seeking, Initiative*** **TR:** AFMAN 33-396, AFMAN 36-2234, CJCSI 3500.01, DAFI 36-2691, DAFMAN 36-2100, DAFMAN 36-2689												
1.5.1. TRAINING (BASIC) **Required Behaviors:** **1. Completes training to meet organizational requirements** **2. Applies and utilizes training resources to increase individual skills and abilities** **3. Demonstrates basic proficiency for practical application** **4. Identifies barriers, issues, and concerns to inform updates** **Criteria:** ***Reach of Influence: Individuals***												
1.5.1.1. Identify training requirements	1	*							-	**K**	-	-
1.5.1.2. Follow individual training plan	1	*							-	**P**	-	-
1.5.1.3. List training resources available to assigned position	2	*							-	**K**	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.5.1.4. Demonstrate how to access and use training materials	2	*							-	P	-	-
1.5.1.5. Identify standard procedures for executing tasks	3	*							-	K	-	-
1.5.1.6. Execute basic skills of assigned position in a supervised or simulated environment	3	*							-	P	-	-
1.5.1.7. Identify individual training challenges that may affect completion or comprehension	4	*							-	K	-	-
1.5.1.8. Describe the appropriate method for reporting training concerns or recommending improvements	4	*							-	K	-	-
1.5.2. TRAINING (INTERMEDIATE) **Required Behaviors:** **1. Reviews and updates training records to document progress** **2. Seeks training opportunities to enhance mission execution capabilities** **3. Observes and provides feedback to assess trainees' proficiency** **4. Trains and develops members in accordance with requirements to enable mission execution** **Criteria:** ***Reach of Influence: Unit/Groups***												
1.5.2.1. Update training records	1	*							-	-	P	-
1.5.2.2. Analyze training data to identify trends, gaps, or overdue requirements	1								-	-	-	-
1.5.2.3. Identify relevant courses, events, and training opportunities	2	*							-	-	K	-
1.5.2.4. Evaluate how supplemental training contributes to increased mission effectiveness	2								-	-	-	-
1.5.2.5. Conduct trainee performance checks	3								-	-	-	-
1.5.2.6. Give feedback to reinforce desired behaviors	3								-	-	-	-
1.5.2.7. Execute training plans and development activities	4	*							-	-	pk	-
1.5.2.8. Adapt training techniques to meet the needs of individuals or teams	4								-	-	-	-
1.5.3. TRAINING (ADVANCED) **Required Behaviors:** **1. Develops and approves instructional material in accordance with requirements to develop skills, knowledge and abilities** **2. Provides foundational and advanced training to meet developmental milestones** **3. Assesses and updates training programs to ensure relevance and effectiveness** **Criteria:** ***Reach of Influence: Wing/Institutional***												
1.5.3.1. Design curriculum	1								-	-	-	-
1.5.3.2. Evaluate curriculum	1								-	-	-	-
1.5.3.3. Field structured training	2								-	-	-	-
1.5.3.4. Tailor instructional environment	2								-	-	-	-
1.5.3.5. Conduct systematic evaluations of training programs	3								-	-	-	-
1.5.3.6. Shape the training requirements development process (e.g. personnel, equipment, resources)	3								-	-	-	-
1.5.3.7. Implement emerging capabilities in training systems	3								-	-	-	-
1.5.3.8. Advocate for training requirements	3								-	-	-	-
1.5.4. TRAINING (EXPERT) **Required Behaviors:** **1. Establishes and maintains life-cycle training requirements to ensure career development** **2. Advocates and coordinates opportunities to achieve and maintain occupational relevancy** **3. Conducts mission analysis to determine training requirements** **Criteria:** ***Reach of Influence: MAJCOM/AF-Level/Industry***												
1.5.4.1. Design life-cycle training frameworks	1								-	-	-	-
1.5.4.2. Institutionalize life-cycle training frameworks	1								-	-	-	-
1.5.4.3. Oversee training strategies	1								-	-	-	-
1.5.4.4. Advocate for institutional and interagency partnerships	2								-	-	-	-
1.5.4.5. Shape force design	2								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
1.5.4.6. Synthesize mission needs	3								-	-	-	-
1.5.4.7. Translate requirements into curriculum development, resource allocation, and strategic readiness	3								-	-	-	-
1.5.4.8. Validate training requirements	3								-	-	-	-
2. AIRMANSHIP												
2.1. PERSONNEL MANAGEMENT **The management of people through deliberate planning, institutional training, development, direction, and organization to maximize performance.** **Supporting Competencies:** ***Accountability, Communication, Develops People, Leadership*** **TR:** AFH 1-1 Ch 13, DAFI 38-101, DAFMAN 17-1305, DAFMAN 36-2905, DAFMAN 38-208												
2.1.1. PERSONNEL MANAGEMENT (BASIC) **Required Behaviors:** **1. Adheres to documented expectations and seeks feedback to ensure compliance to standards** **2. Manages tasks to accomplish desired actions or outcomes** **3. Articulates responsibilities, impact, and role to demonstrate understanding of position** **4. Identify organizational structure models to demonstrate understanding of force design** **Criteria:** ***Scope: Specific Area***												
2.1.1.1. Identify feedback mechanisms	1	*							-	**K**	-	-
2.1.1.2. Identify Department of the Air Force policy regarding standards (e.g. Dress and Appearance, Airman Comprehensive Assessment, Fitness, etc.)	1	*							-	**K**	-	-
2.1.1.3. Explain responsibilities of ratee and rater roles	1	*							-	**K**	-	-
2.1.1.4. Translate assigned tasks into an executable plan	2	*							-	**K**	-	-
2.1.1.5. Evaluate results against a plan's desired outcome	2	*							-	**pk**	-	-
2.1.1.6. Describe positional impact on the mission	3	*							-	**K**	-	-
2.1.1.7. Explain function of individual position within organization	3	*							-	**K**	-	-
2.1.1.8. Describe AFSC structure and composition	4	*							**K**	-	-	-
2.1.1.9. Distinguish between administrative and operational chains of command	4	*							**K**	-	-	-
2.1.2. PERSONNEL MANAGEMENT (INTERMEDIATE) **Required Behaviors:** **1. Documents expectations and feedback to ensure adherence to standards** **2. Establishes subordinates responsibilities, impact, and role to support organizational requirements** **3. Identifies applicable personnel system resources and available products to inform and manage subordinates** **4. Utilizes available tools and programs to highlight individual and team successes** **Criteria:** ***Scope: Integration with concerned areas***												
2.1.2.1. Select appropriate feedback mechanisms	1								-	-	-	-
2.1.2.2. Conduct feedback sessions	1								-	-	-	-
2.1.2.3. Communicate changes in standards	1	*							-	-	**K**	-
2.1.2.4. Enforce standards	1								-	-	-	-
2.1.2.5. Set expectations	2								-	-	-	-
2.1.2.6. Explain the relationship between unit mission and national objectives	2	*							-	**K**	-	-
2.1.2.7. Explain mission impact of subordinate's role	2								-	-	-	-
2.1.2.8. Identify available personnel management products and systems	3								-	-	-	-
2.1.2.9. Use available personnel management product and systems	3								-	-	-	-
2.1.2.10. Leverage performance reporting and decoration programs	4								-	-	-	-
2.1.2.11. Recognize high performers using available programs	4								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
2.1.3. PERSONNEL MANAGEMENT (ADVANCED) **Required Behaviors:** **1. Establishes policies and manages administrative requirements to enhance effectiveness** **2. Identifies personnel constraints and proposes solutions to solve deficiencies** **3. Provides feedback and vectors personnel based on performance to inform career advancement (e.g. DT Process)** **Criteria:** ***Scope: Integration with organizational strategies***												
2.1.3.1. Analyze gaps in organizational effectiveness	1								-	-	-	-
2.1.3.2. Develop policy that addresses gaps in organizational effectiveness	1								-	-	-	-
2.1.3.3. Assess manpower shortfalls	1	*							-	-	-	**K**
2.1.3.4. Leverage processes to address shortfalls	2								-	-	-	-
2.1.3.5. Create development plans	2	*							-	-	-	**pk**
2.1.3.6. Influence force structure	2								-	-	-	-
2.1.3.7. Execute development panel (e.g. EFDP/DTs)	3								-	-	-	-
2.1.3.8. Evaluate feedback mechanisms effectiveness	3								-	-	-	-
2.1.3.9. Manage manpower requirements (e.g. UMD/PGL/TPR)	3	*							-	-	-	**pk**
2.1.4. PERSONNEL MANAGEMENT (EXPERT) **Required Behaviors:** **1. Formulates strategy to manage manpower** **2. Manages requisitions authorizations and coordinates change requests to optimize organizational requirements** **Criteria:** ***Scope: Integration with AF-level/within industry***												
2.1.4.1. Develop Talent Management Framework(s)	1								-	-	-	-
2.1.4.2. Identify strategic manpower impacts	1								-	-	-	-
2.1.4.3. Determine key leadership positions	1								-	-	-	-
2.1.4.4. Evaluate career field manpower structure/force design	1								-	-	-	-
2.1.4.5. Lead manpower study assessments	1								-	-	-	-
2.1.4.6. Validate manpower change requests	2								-	-	-	-
2.1.4.7. Validate organizational change requests	2								-	-	-	-
2.1.4.8. Evaluate organizational design	2								-	-	-	-
2.1.4.9. Validate requests for career field manpower	2								-	-	-	-
2.2. PROGRAM MANAGEMENT **The development, implementation, and validation of administrative programs to enhance and sustain organizational performance.** **Supporting Competencies: Accountability, Decision Making, Organizational Management, Resource Management** **TR:** AFH 1-1 Ch 11, DAFI 38-401, DAFI 90-201, DAFMAN 17-130, DAFMAN 90-203												
2.2.1. PROGRAM MANAGEMENT (BASIC) **Required Behaviors:** **1. Demonstrates familiarization with programs and applicability to ensure individual administrative readiness/Demonstrate familiarization with commander's programs to ensure compliance** **2. Follows established procedures, policy, and guidance to meet administrative program requirements** **Criteria: Scope: Specific Area**												
2.2.1.1. Identify commander's programs	1	*							**K**	-	-	-
2.2.1.2. Identify programs' purpose, roles, and responsibilities	1	*							**K**	-	-	-
2.2.1.3. Identify program standardization and guidance	2	*							**K**	-	-	-
2.2.1.4. Adhere to required timelines	2								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
2.2.1.5. Identify impact of guidance and policy	2	*							K	-	-	-
2.2.1.6. Identify program resources	2								-	-	-	-
2.2.2. PROGRAM MANAGEMENT (INTERMEDIATE) **Required Behaviors:** **1. Educates junior members on the applicability of administrative programs to ensure familiarization** **2. Serves as alternate administrator to provide program continuity** **3. Utilizes digital tools and platforms to implement programs** **4. Implements continuous improvement processes to optimize program effectiveness** **Criteria: Scope: Integration with concerned areas**												
2.2.2.1. Explain importance of administrative programs and impact to mission	1								-	-	-	-
2.2.2.2. Utilize resources	1								-	-	-	-
2.2.2.3. Explain program roles and responsibilities	2								-	-	-	-
2.2.2.4. Execute applicable role responsibilities	2	*							pk	-	-	-
2.2.2.5. Identify tools and resources	3								-	-	-	-
2.2.2.6. Utilize tools and resources	3								-	-	-	-
2.2.2.7. Explain continuous improvement processes	4								-	-	-	-
2.2.2.8. Implement continuous improvement processes	4								-	-	-	-
2.2.3. PROGRAM MANAGEMENT (ADVANCED) **Required Behaviors:** **1. Serves as program administrator to ensure compliance** **2. Communicates on program implementation, compliance and gaps to inform decision-making** **3. Mitigate program risks to reduce negative mission impact** **4. Develop program continuity guidance to sustain commander's program** **Criteria: Scope: Integration with organizational strategies**												
2.2.3.1. Execute program requirements	1								-	-	-	-
2.2.3.2. Evaluate program compliance	1								-	-	-	-
2.2.3.3. Brief program status	2								-	-	-	-
2.2.3.4. Document program compliance	2								-	-	-	-
2.2.3.5. Analyze program risks	3								-	-	-	-
2.2.3.6. Develop risk mitigation strategy	3								-	-	-	-
2.2.3.7. Implement risk mitigation strategy	3								-	-	-	-
2.2.3.8. Develop continuity procedures	4								-	-	-	-
2.2.3.9. Develop emergency protocols and procedures	4								-	-	-	-
2.2.4. PROGRAM MANAGEMENT (EXPERT) **Required Behaviors:** **1. Provides guidance on program implementation to ensure standards compliance** **2. Makes decisions and takes actions to ensure successful program implementation** **3. Ensures program objectives are aligned to policy intent** **Criteria:** ***Scope: Integration with AF-level/within industry***												
2.2.4.1. Establish program requirements	1								-	-	-	-
2.2.4.2. Develop inspection criteria	1								-	-	-	-
2.2.4.3. Validate program compliance	2								-	-	-	-
2.2.4.4. Assess program and policy	2								-	-	-	-
2.2.4.5. Interpret policy intent	3								-	-	-	-
2.2.4.6. Update program	3								-	-	-	-
2.2.4.7. Advise on policy changes	3								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
2.3. RESOURCE MANAGEMENT **The allocation, utilization, and oversight of resources to ensure mission readiness and operational effectiveness.** **Supporting Competencies:** ***Accountability, Decision Making, Organizational Awareness, Resource Management*** **TR:** AFH 1-1 Ch12-13, AFI 17-213, AFI 65-103, AFI 65-601V2, AFMAN 65-604, CJCSI 5123.01, DAFI 23-111, DAFI 64-117, DAFI 65-601V1												
2.3.1. RESOURCE MANAGEMENT (BASIC) **Required Behaviors:** **1. Identifies resources required to perform basic unit tasks** **2. Identifies resourcing shortfalls to inform leadership situational awareness** **Criteria:** ***Impact on… Specific workplace tasks***												
2.3.1.1. List resources needed	1	*							-	**K**	-	-
2.3.1.2. Select appropriate tools	1	*							-	**pk**	-	-
2.3.1.3. Utilize resources	1								-	-	-	-
2.3.1.4. Recognize resource shortfalls	2	*							-	**K**	-	-
2.3.1.5. Report shortages	2	*							-	**pk**	-	-
2.3.1.6. Explain impact to mission	2	*							-	**K**	-	-
2.3.2. RESOURCE MANAGEMENT (INTERMEDIATE) **Required Behaviors:** **1. Assesses unit projects for resourcing needs to ensure successful project completion** **2. Articulates resourcing shortfalls to ensure successful project completion** **Criteria:** ***Impact on… Specific workplace projects***												
2.3.2.1. Evaluate project resource requirements	1								-	-	-	-
2.3.2.2. Forecast future project resource needs	1								-	-	-	-
2.3.2.3. Report resource efficiency	1								-	-	-	-
2.3.2.4. Evaluate resource inventory	2								-	-	-	-
2.3.2.5. Develop reporting metrics	2								-	-	-	-
2.3.2.6. Deliver shortfall solutions	2								-	-	-	-
2.3.3. RESOURCE MANAGEMENT (ADVANCED) **Required Behaviors:** **1. Evaluates unit projects to determine required resources** **2. Secures unit-level resources for project completion** **3. Provides impact to unit leadership to mitigate resource shortfalls** **Criteria:** ***Impact on… Management decisions***												
2.3.3.1. Evaluate resource requirements	1								-	-	-	-
2.3.3.2. Justify resource requirements	1								-	-	-	-
2.3.3.3. Influence resource decisions	2								-	-	-	-
2.3.3.4. Advocate for resources	2								-	-	-	-
2.3.3.5. Identify resource gaps	3								-	-	-	-
2.3.3.6. Develop risk assessments	3								-	-	-	-
2.3.3.7. Report resourcing shortfalls	3								-	-	-	-
2.3.4. RESOURCE MANAGEMENT (EXPERT) **Required Behaviors:** **1. Oversees projects to ensure required resources are available for execution** **2. Submits documentation to support resourcing requirements** **Criteria:** ***Impact on… AF-level practices/within industry***												
2.3.4.1. Validate requirements	1								-	-	-	-
2.3.4.2. Develop requirements	1								-	-	-	-
2.3.4.3. Develop resource management policy and procedures	2								-	-	-	-
2.3.4.4. Align resources to mission requirements	2								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
2.3.4.5. Determine resource requirements based on emerging mission needs	2								-	-	-	-
2.3.4.6. Advocate for priorities	2								-	-	-	-
3. MISSION EXECUTION												
3.1. ELECTROMAGNETIC SPECTRUM **The aggregate of knowledge and application of technologies employing electromagnetic radiation, signals and transmissions, tradecraft, and production of effects.** **Supporting Competencies:** ***Analytical Thinking, Creative Thinking, Digital Literacy, Information Seeking*** **TR:** AFTTP 3-1, AFTTP 3-3 C2, DAFMAN 10-703, DAFPD 10-15, JP 3-52 Ch 3-c, JP 3-85, NIST SP 800-153												
3.1.1. ELECTROMAGNETIC SPECTRUM (BASIC) **Required Behaviors:** **1. Determines the frequencies, channels, transmission rates, and bandwidth of signals of interest to characterize activity** **2. Utilizes tools/equipment for EMS actions** **3. Explains the interaction of emissions within the environment to support mission execution** **Criteria:** ***Depth of Knowledge: Established practice with some workplace elements***												
3.1.1.1. Define basic and operational terms (e.g., frequency, wavelength, ,propagation, etc.)	1	*							**K**	-	-	-
3.1.1.2. List primary frequency ranges of the electromagnetic spectrum (e.g., LF, VHF, UHF, etc.)	1	*							**K**	-	-	-
3.1.1.3. Match frequency bands with their common military or civilian applications (e.g., Wi-Fi, AM/FM Radio, SATCOM, IIoT, OT etc.)	1	*							**K**	-	-	-
3.1.1.4. Demonstrate ability to read/interpret spectrogram	1	*							**P**	-	-	-
3.1.1.5. Identify common tools associated with EMS operation (e.g., GNU Radio, wire shark, collect/replay toolkits, etc.)	2	*							**K**	-	-	-
3.1.1.6. Identify common equipment associated with EMS operation (e.g., spectrum analyzer, SDR, signal generators, SATCOM terminals, tactical radios, antennas, amplifiers, etc.)	2	*							**K**	-	-	-
3.1.1.7. Identify common equipment configurations for EMS capabilities	2	*							**K**	-	-	-
3.1.1.8. Utilize common tools associated with EMS operation (GNU Radio, Wireshark, collect/replay toolkits, etc.)	2	*							**P**	-	-	-
3.1.1.9. Utilize common equipment associated with EMS operation (spectrum analyzer, SDR, signal generators, SATCOM terminals, tactical radios, antennas, amplifiers, etc.)	2	*							**P**	-	-	-
3.1.1.10. Describe basic functions of EMS equipment	2	*							**K**	-	-	-
3.1.1.11. Explain wave propagation characteristics	3	*							**K**	-	-	-
3.1.1.12. List environments and environmental conditions that have an impact on emissions	3	*							**K**	-	-	-
3.1.1.13. Identify different material effects on various EMS frequencies	3	*							**K**	-	-	-
3.1.1.14. Describe architectures, protocols, and vulnerabilities of wireless and mobile technologies	3	*							**K**			

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.1.2. ELECTROMAGNETIC SPECTRUM (INTERMEDIATE) **Required Behaviors:** **1. Use and explain available frequencies within the environment to gain tactical advantage** **2. Extracts, generates, or inhibits signals appropriately for effective execution of tactical plans** **3. Selects, configures, and troubleshoots equipment needed to interact with the EMS** **Criteria:** ***Depth of Knowledge: Established practices of all workplace elements***												
3.1.2.1. Analyze survey to determine RF operational environment	1	*							**P**	-	-	-
3.1.2.2. Deconflict frequency band usage to ensure spectrum availability	1	*							**P**	-	-	-
3.1.2.3. Demonstrate ability to maneuver within RF environment	2	*							**P**	-	-	-
3.1.2.4. Identify opportunities for spectrum cohabitation	2								-	-	-	-
3.1.2.5. Utilize tools to locate/collect signals of interest	2	*							**P**	-	-	-
3.1.2.6. Select appropriate antenna types and placements for given frequency bands and mission goals	3	*							**K**	-	-	-
3.1.2.7. Troubleshoot issues in capturing/sending data	3	*							**P**	-	-	-
3.1.2.8. Validate performance metrics of EMS tools (noise floor, SNR, bandwidth, etc.)	3	*							**P**	-	-	-
3.1.3. ELECTROMAGNETIC SPECTRUM (ADVANCED) **Required Behaviors:** **1. Develops TTPs based on environmental conditions and actors to achieve mission objectives** **2. Determines detection or subversion methods to impact mission-critical signals** **3. Modifies existing equipment functions, adapting to dynamic mission requirements** **Criteria:** ***Depth of Knowledge: New practices of all workplace elements***												
3.1.3.1. Evaluate EMS mission data, environmental changes, and actor responses	1								-	-	-	-
3.1.3.2. Develop EMS TTPs tailored to specific terrain and propagation models	1								-	-	-	-
3.1.3.3. Analyze target EMS signature and usage	2								-	-	-	-
3.1.3.4. Develop Low Probability of Intercept (LPI)/ Low Probability of Detect (LPD) signal guidance	2								-	-	-	-
3.1.3.5. Demonstrate dynamic transmission power	3								-	-	-	-
3.1.3.6. Engineer field-adaptable RF paths in constrained environments	3								-	-	-	-
3.1.3.7. Refine deployment packages and technical guides	3								-	-	-	-
3.1.4. ELECTROMAGNETIC SPECTRUM (EXPERT) **Required Behaviors:** **1. Identifies, creates, or integrates emerging capabilities to achieve overmatch within the environment** **2. Provides technical expertise to advise stakeholders** **Criteria:** ***Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area***												
3.1.4.1. Identify utilization of protocols outside of standard spectrum range	1								-	-	-	-
3.1.4.2. Develop unconventional RF configurations using non-standard power, control, and data interfaces	1								-	-	-	-
3.1.4.3. Validate unconventional RF configurations using non-standard power, control, and data interfaces	1								-	-	-	-
3.1.4.4. Develop advanced signal manipulation capabilities	1								-	-	-	-
3.1.4.5. Validate advanced signal manipulation capabilities	1								-	-	-	-
3.1.4.6. Identify emerging technical trends	1								-	-	-	-
3.1.4.7. Inform acquisition of equipment and training with key mission partners and industry	2								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.1.4.8. Develop doctrine and training material on advanced EMS employment	2								-	-	-	-
3.1.4.9. Evaluate field suitability of emerging EMS capabilities	2								-	-	-	-
3.1.4.10. Guide integration of emerging EMS capabilities	2								-	-	-	-
3.2. HARDWARE **The aggregate of knowledge and application of technologies employing tangible components and firmware, defining infrastructure across physical and virtualized levels.** **Supporting Competencies:** ***Analytical Thinking, Creative Thinking, Digital Literacy, Information Seeking*** **TR:** AFMAN 17-1203, JP 3-12, JP 3-85, JP 3-52 Ch 3-c, NIST SP 800-125A, NIST SP 800-193, NIST SP 800-82												
3.2.1. HARDWARE (BASIC) **Required Behaviors:** **1. Describes hardware components to articulate functional and mission purpose** **2. Follows specified hardware maintenance procedures to ensure operational function** **3. Identifies hardware states to determine troubleshooting requirements** **4. Explains components of infrastructures, physical and virtual, for different use cases** **Criteria:** ***Depth of Knowledge: Established practice with some workplace elements***												
3.2.1.1. Define hardware terminology (e.g., Information Technology (IT), Operational Technology (OT), Platform Information Technology (PIT)	1	*							**K**	-	-	-
3.2.1.2. Describe hardware functionality	1	*							**K**	-	-	-
3.2.1.3. Differentiate between Information Technology (IT) and Operational Technology (OT)	1	*							**K**	-	-	-
3.2.1.4. Identify internal computing components of IT/OT hardware	1	*							**K**	-	-	-
3.2.1.5. Describe how computing components interact	1	*							**K**	-	-	-
3.2.1.6. Identify hardware limitations	1	*							**K**	-	-	-
3.2.1.7. Identify key physical terrain	1	*							**K**	-	-	-
3.2.1.8. Review technical guidance	2	*							**K**	-	-	-
3.2.1.9. Recognize hardware limitations within physical and digital operating environments	2								-	-	-	-
3.2.1.10. Explain resourcing requirements	2								-	-	-	-
3.2.1.11. Recognize indicators of failure	3								-	-	-	-
3.2.1.12. Differentiate between hardware states	3								-	-	-	-
3.2.1.13. Demonstrate safeguarding methods of securing hardware	4								-	-	-	-
3.2.1.14. Explain the different IT/OT infrastructures and networking architecture	4	*							**K**	-	-	-
3.2.1.15. Recognize different control systems	4								-	-	-	-
3.2.1.16. Identify supporting hardware infrastructure and limiting conditions to mission assurance (loss of capabilities)	4	*							**K**	-	-	-
3.2.2. HARDWARE (INTERMEDIATE) **Required Behaviors:** **1. Describes hardware components to articulate functional and mission purpose** **2. Follows specified hardware maintenance procedures to ensure operational function** **3. Identifies hardware states to determine troubleshooting requirements** **4. Explains components of infrastructures, physical and virtual, for different use cases** **Criteria:** ***Depth of Knowledge: Established practice with some workplace elements***												
3.2.1.1. Install hardware components	1								-	-	-	-
3.2.1.2. Update firmware	1								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT					4. Proficiency Codes Used To			
				A	B	C	D	E	A	B	C	D
				Tng Start	Tng Complete	Trainee Initials	Trainer Initials	Certifier Initials	3-lvl IST	5-lvl A1C	7-lvl SSgt	9-lvl SMSgt
3.2.1.2. Demonstrate the ability to find and manipulate common hardware sockets	1								-	-	-	-
3.2.1.3. Scale hardware for environments	2								-	-	-	-
3.2.1.4. Modify virtual hardware system resources	2								-	-	-	-
3.2.1.5. Differentiate the limitations and capabilities of various hardware components	3								-	-	-	-
3.2.1.6. Evaluate aggregate hardware capabilities	3								-	-	-	-
3.2.1.7. Explain supporting infrastructure and environmental limitations	3								-	-	-	-
3.2.1.8. Analyze firmware	4								-	-	-	-
3.2.1.9. Troubleshoot hardware and firmware	4								-	-	-	-
3.2.1.10. Determine impacts from hardware changes	4								-	-	-	-
3.2.3. HARDWARE (ADVANCED) **Required Behaviors:** **1. Identifies, obtains, and integrates appropriate hardware solutions to meet mission needs** **2. Leverages or incorporates available external infrastructure to enhance mission effectiveness** **3. Seeks, architects, and implements optimizations to redistribute hardware workloads across mission infrastructure** **4. Enhances, repurposes, or changes firmware functionality to meet mission needs** **Criteria:** ***Depth of Knowledge: Established practices of all workplace elements***												
3.2.3.1. Prioritize hardware solutions and capabilities	1								-	-	-	-
3.2.3.2. Plan procurement and implementation timelines	1								-	-	-	-
3.2.3.3. Validate procurement requirements	1								-	-	-	-
3.2.3.4. Integrate external capabilities	2								-	-	-	-
3.2.3.5. Shape requirements for external capabilities	2								-	-	-	-
3.2.3.6. Optimize hardware configuration	3								-	-	-	-
3.2.3.7. Design scalable architectures	3								-	-	-	-
3.2.3.8. Demonstrate knowledge of Supervisory Control and Data Acquisition (SCADA) systems, Program Logic Controllers (PLCs), Human-Machine Interface (HMIs), and the criticality of maintaining operational stability	3								-	-	-	-
3.2.3.9. Develop new firmware capabilities	4								-	-	-	-
3.2.3.10. Reverse-Engineer firmware	4								-	-	-	-
3.2.4. HARDWARE (EXPERT) **Required Behaviors:** **1. Creates or integrates emerging firmware capabilities to develop solutions** **2. Envisions, engineers, and fabricates purpose-specific hardware solutions to maintain competitive advantage** **3. Provides technical expertise to advise stakeholders** **Criteria:** ***Depth of Knowledge: New practices of all workplace elements***												
3.2.4.1. Craft non-traditional capabilities	1								-	-	-	-
3.2.4.2. Integrate current capabilities with emerging IT/OT technology	1								-	-	-	-
3.2.4.3. Evaluates emerging firmware architectures	1								-	-	-	-
3.2.4.5. Develop R&D partnerships	2								-	-	-	-
3.2.4.6. Develop hardware to bring capabilities to non-traditional areas	2								-	-	-	-
3.2.4.7. Advocate to stakeholders	3								-	-	-	-
3.2.4.8. Characterize hardware risks	3								-	-	-	-
3.3. NETWORKING **Description: The aggregate of knowledge and technologies employing the logical interconnection of devices to enable information transfer between systems.** **Sub-Competencies:** ***Analytical Thinking, Creative Thinking, Digital Literacy, Information Seeking*** **TR:** Cisco CCNA/CCENT; IEEE 802, MITRE ATT&CK Framework, NIST SP 800-82												

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT					4. Proficiency Codes Used To			
				A	B	C	D	E	A	B	C	D
				Tng Start	Tng Complete	Trainee Initials	Trainer Initials	Certifier Initials	3-lvl IST	5-lvl A1C	7-lvl SSgt	9-lvl SMSgt
3.3.1. NETWORKING (BASIC) **Required Behaviors:** **1. Selects and uses capabilities to characterize network activity** **2. Identify network behavior to characterize anomalies** **3. Develops and interprets network maps to define the environment** **4. Describes network architectures to inform mission needs** **Criteria: Depth of Knowledge: Established practice with some workplace elements**												
3.3.1.1. Identify Operational Technology (OT) ports, protocols, traffic flows of different OT vendor systems	1	*							**K**	-	-	-
3.3.1.2. Utilize capabilities to observe and capture network activity	1	*							**P**	-	-	-
3.3.1.3. Identify the function and purpose of captured network activity	1	*							**K**	-	-	-
3.3.1.4. Extract data from captured network activity	1	*							**P**	-	-	-
3.3.1.5. Describe network data structures	1	*							**K**	-	-	-
3.3.1.6. Describe common network-based exploits and associated indicators of compromise	2	*							**K**	-	-	-
3.3.1.7. Describe protections and mitigations against common network-based exploits	2	*							**K**	-	-	-
3.3.1.8. Identify network devices	3	*							**K**	-	-	-
3.3.1.9. Explain network device functionality	3	*							**K**	-	-	-
3.3.1.10. Interpret physical and logical network maps	3	*							**P**	-	-	-
3.3.1.11. Construct network logical and physical topology diagrams	3	*							**P**	-	-	-
3.3.1.12. Summarize the difference between cloud networking and cloud computing	4	*							**K**	-	-	-
3.3.1.13. Identify network architectures (IT, OT, PIT) and their typical use cases	4	*							**K**	-	-	-
3.3.1.14. Identify network elements of control systems	4	*							**K**	-	-	-
3.3.2. NETWORKING (INTERMEDIATE) **Required Behaviors:** **1. Configures network capabilities to enhance functionality** **2. Articulates and assesses network activity to support decision making** **3. Selects and determines appropriate locations to employ network capabilities** **Criteria:** ***Depth of Knowledge: Established practices of all workplace elements***												
3.3.2.1. Modify IT, OT and PIT network configurations	1	*							**P**	-	-	-
3.3.2.2. Analyze the impact of configuration changes in IT, OT and PIT	1	*							**P**	-	-	-
3.3.2.3. Implement network segmentation (e.g., Virtual Local Area Networks (VLANs), Virtual Private Networks (VPNs), etc.)	1	*							**P**	-	-	-
3.3.2.4. Characterize modern cryptography protocols and their typical use cases	1	*							**K**	-	-	-
3.3.2.5. Identify anomalies and actor behavior	2	*							**K**	-	-	-
3.3.2.6. Characterize the operational environment	2	*							**P**	-	-	-
3.3.2.7. Extract pattern of life from network traffic	2	*							**P**	-	-	-
3.3.2.8. Conduct vulnerability assessments on IT, OT and PIT	2	*							**P**	-	-	-
3.3.2.9. Communicate details of network activity	2	*							**P**	-	-	-
3.3.2.10. Determine capability placement in IT, OT and PIT	3	*							**P**	-	-	-
3.3.2.11. Deploy capabilities in IT, OT and PIT	3	*							**pk**	-	-	-
3.3.2.12. Operate in, on, and through the cloud	3								-	-	-	-
3.3.2.13. Distinguish remote access methods for IT, OT and PIT	3	*							**K**	-	-	-
3.3.2.14. Articulate security implications of remote access methods	3	*							**K**	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.3.3. NETWORKING (ADVANCED) **Required Behaviors:** **1. Applies techniques to resolve complex networking issues** **2. Uses network capabilities and access methods to automate, optimize, or obfuscate interactions** **3. Develops TTPs adapting to dynamic mission requirements** **4. Modifies, adapts, or integrates existing capabilities to accomplish mission objectives** **Criteria:** ***Depth of Knowledge: New practices of all workplace elements***												
3.3.3.1. Resolve indicators of misconfiguration, degradation, or adversary presence	1								-	-	-	-
3.3.3.2. Enrich physical and logical network maps with collected data	1								-	-	-	-
3.3.3.3. Validate network maps	1								-	-	-	-
3.3.3.4. Employ and detect access methods or communication channels	2								-	-	-	-
3.3.3.5. Employ and detect obfuscation techniques	2	*							**K**	-	-	-
3.3.3.6. Develop capabilities that analyze and modify network traffic (e.g., IT, OT, PIT)	2								-	-	-	-
3.3.3.7. Create playbooks or SOPs	3								-	-	-	-
3.3.3.8. Fuse threat-intelligence from multiple sources	3								-	-	-	-
3.3.3.9. Adapt advanced network architectures	4								-	-	-	-
3.3.3.10. Integrate multiple capabilities into a common operational picture	4								-	-	-	-
3.3.3.11. Harden IT, OT and PIT environments	4								-	-	-	-
3.3.4. NETWORKING (EXPERT) **Required Behaviors:** **1. Provides technical expertise to advise stakeholders** **2. Collect, characterizes, and affects proprietary or specialized network data to support mission objectives** **3. Envisions, advocates for, or engineers network architecture to optimize or achieve overmatch** **Criteria:** ***Depth of Knowledge: New practices of all workplace elements***												
3.3.4.1. Advise stakeholders	1								-	-	-	-
3.3.4.2. Evaluate emerging technologies	1								-	-	-	-
3.3.4.3. Collect specialized network data	2								-	-	-	-
3.3.4.4. Analyze specialized data, protocols, and devices	2								-	-	-	-
3.3.4.5. Research specialized protocols	2								-	-	-	-
3.3.4.6. Develop partnerships	3								-	-	-	-
3.3.4.7. Develop unique solutions to overcome capability limitations	3								-	-	-	-
3.3.4.8. Formulate new concepts and architectural frameworks	3								-	-	-	-
3.3.4.9. Integrate emerging network technologies and methodologies	3								-	-	-	-
3.3.4.10. Develop emerging requirements	3								-	-	-	-
3.3.4.11. Author guidance	3								-	-	-	-
3.4. SOFTWARE **Description: The aggregate of knowledge and technologies to develop the logical instruction sets, data, or programs that tell a system what to do and how to perform specific tasks.** **Supporting Competencies:** ***Analytical Thinking, Creative Thinking, Digital Literacy, Information Seeking*** **TR:** AF e-Learning , FedVTE: Windows Operating System Security, JP 3-12, Linux: Basic System Administration Learning Track (Percipio & FedVTE); UNIX Essentials Learning Track (Percipio & FedVTE), NIST SP 800-86												

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.4.1. SOFTWARE (BASIC) **Required Behaviors:** **1. Articulates software structures to characterize functionality** **2. Builds or troubleshoots scripts and programs to complete tasks** **3. Identifies concepts (e.g. code and security) across various languages, computer architectures, and platforms to define the environment** **4. Explains the orchestration of system processes alongside component integration to achieve mission requirements** **Criteria:** ***Depth of Knowledge: Established practice with some workplace elements***												
3.4.1.1. Describe the difference between interpreted languages and compiled languages	1	*							**K**	-	-	-
3.4.1.2. Identify purpose and functionality of an existing script	1	*							**K**	-	-	-
3.4.1.3. Recognize common file extensions associated with scripts and executable programs	1	*							**K**	-	-	-
3.4.1.4. Describe the general purpose of software libraries	1	*							**K**	-	-	-
3.4.1.5. Identify software structures	1	*							**K**	-	-	-
3.4.1.6. Execute fundamental computing concepts, numbering systems, and logical operations	1	*							**P**			
3.4.1.7. Execute operating system commands and scripts on local and remote systems	2	*							**P**	-	-	-
3.4.1.8. Modify existing scripts	2	*							**P**	-	-	-
3.4.1.9. Troubleshoot common error messages	2	*							**P**	-	-	-
3.4.1.10. Develop and execute basic programs or scripts	2	*							**P**			
3.4.1.11. Apply encryption, encoding, and obfuscation techniques to data and code	2	*							**P**			
3.4.1.12. Identify core operating system components and their fundamental roles in various environments	3	*							**K**	-	-	-
3.4.1.13. Recognize computer security concepts	3	*							**K**	-	-	-
3.4.1.14. Describe the concept of virtualization and its common applications	3	*							**K**	-	-	-
3.4.1.15. Distinguish a virtualized operating environment from a physical one	3	*							**P**	-	-	-
3.4.1.16. Differentiate between local and remote software execution	3	*							**K**	-	-	-
3.4.1.17. Identify common characteristics and use cases of Operational Technologies (OT)	3	*							**K**	-	-	-
3.4.1.18. Explain the relationship between software, firmware, and hardware	3	*							**K**	-	-	-
3.4.1.19. Differentiate between system architectures	3	*							**K**	-	-	-
3.4.1.20. Describe how a tool or script utilizes operating system resources	4	*							**K**	-	-	-
3.4.1.21. Describe dependencies between different software components and/or system services	4	*							**K**	-	-	-
3.4.1.22. Identify how AI/LLM tools can be applied	4	*							**K**	-	-	-
3.4.1.23. Manage VMs and containers	4	*							**pk**	-	-	-
3.4.2. SOFTWARE (INTERMEDIATE) **Required Behaviors:** **1. Analyzes and configures a script or program to mitigate vulnerabilities or modify functionality** **2. Articulates and assesses system activity to support decision making** **3. Selects and determines appropriate locations to employ system or software capabilities** **4. Develop, configures, and troubleshoots solutions with existing capabilities to control operating environment** **Criteria:** ***Depth of Knowledge: Established practices of all workplace elements***												
3.4.2.1. Mitigate vulnerabilities	1	*							**K**	-	-	-
3.4.2.2. Modify capabilities	1								-	-	-	-
3.4.2.3. Describe how capabilities utilize specific software libraries, modules, or APIs	1	*							**K**	-	-	-
3.4.2.4. Identify patterns of activity and potential anomalies	2	*							**K**	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.4.2.5. Triage observed system behaviors	2	*							P	-	-	-
3.4.2.6. Articulate the transformation of high-level code to machine code	2								-	-	-	-
3.4.2.7. Articulate assessments of ongoing activity	2	*							K	-	-	-
3.4.2.8. Select appropriate capability	3								-	-	-	-
3.4.2.9. Determine suitable locations for employing capabilities	3								-	-	-	-
3.4.2.10. Identify dependencies and limitations	3								-	-	-	-
3.4.2.11. Identify OT considerations when employing software	3								-	-	-	-
3.4.2.12. Determine target vectors to employ offensive software capabilities	3	*							P			
3.4.2.13. Develop capabilities	4	*							P	-	-	-
3.4.2.14. Deploy capabilities	4	*							P	-	-	-
3.4.2.15. Test functionality	4	*							P	-	-	-
3.4.2.16. Determine impact of capabilities	4	*							P	-	-	-
3.4.2.17. Employ AI/LLM tools	4								-	-	-	-
3.4.2.18. Develop test environments	4								-	-	-	-
3.4.2.19. Validate capabilities	4	*							K	-	-	-
3.4.3. SOFTWARE (ADVANCED) **Required Behaviors:** **1. Develops or reverse-engineers capabilities to support the mission** **2. Deploys system capabilities and access methods to automate, optimize, or obfuscate interactions** **3. Seeks, architects, and implements optimizations to adapt software for complex scenarios** **4. Develops TTPs adapting to dynamic mission requirements** ***Criteria: Depth of Knowledge: New practices of all workplace elements***												
3.4.3.1. Develop capabilities	1								-	-	-	-
3.4.3.2. Reverse engineer capabilities	1								-	-	-	-
3.4.3.3. Conduct forensics	1								-	-	-	-
3.4.3.4. Analyze failures of capabilities	1								-	-	-	-
3.4.3.5.Drive the integration of AI/LLM capabilities	1								-	-	-	-
3.4.3.6. Develop Operational Capabilities (OT) capabilities	1								-	-	-	-
3.4.3.7. Implement multi-stage capabilities	2								-	-	-	-
3.4.3.8. Develop access methods	2								-	-	-	-
3.4.3.9. Optimize capabilities for data transmission and/or system resource utilization	3								-	-	-	-
3.4.3.10. Prototype capabilities	3								-	-	-	-
3.4.3.11. Integrate capabilities	3								-	-	-	-
3.4.3.12. Develop best practices	4								-	-	-	-
3.4.3.13. Develop new TTPs	4								-	-	-	-
3.4.4. SOFTWARE (EXPERT) **Required Behaviors:** **1. Evaluates, creates, integrates emerging or novel concepts and TTPs to improve mission efficiency, accuracy, and adaptability** **2. Creates, innovates, or modifies software and platforms beyond the engineered intent** **3. Provides technical expertise to advise stakeholders** **Criteria:** ***New practices/concepts and theories of all workplace elements; is a credible resource in this area***												
3.4.4.1. Direct the integration of novel concepts or emerging technologies	1								-	-	-	-
3.4.4.2. Institutionalize new TTPs or conceptual frameworks	1								-	-	-	-
3.4.4.3. Re-engineer capabilities	2								-	-	-	-
3.4.4.4. Lead Operational Technology (OT) capability adaptation	2								-	-	-	-
3.4.4.5. Architect capabilities	2								-	-	-	-
3.4.4.6. Advise stakeholders	3								-	-	-	-
3.4.4.7. Author guidance	3								-	-	-	-
3.4.4.8. Advocate for capabilities	3								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.5. ANALYSIS **Description: The use of methods and technologies to conduct data analysis (collect, clean, compile, visualize, present, evaluate) to produce information.** **Sub-Competencies:** ***Analytical, Data Interpretation, Digital, Information Seeking*** **TR:** AFH 1-1 Ch 16, AFMAN 33-396, JP 2-0, JP 2-01, JP 3-04, JP 3-12, JP 3-60												
3.5.1. ANALYSIS (BASIC) **Required Behaviors:** **1. Establish data flows to process information in support of analysis** **2. Queries or scans for mission-related information or data** **3. Applies analytical methods on data and information to draw conclusions** **4. Describes analytical conclusions to stakeholders** **Criteria:** ***Depth of Knowledge: Established practice with some workplace elements***												
3.5.1.1. Identify data and information requirements	1	*							**K**	-	-	-
3.5.1.2. Determine data-flow guiding documentation	1	*							**pk**	-	-	-
3.5.1.3. Describe relationships, data flows, and paths using visualization	1	*							**K**	-	-	-
3.5.1.4. Determine how data flows within the operational environment	1	*							**K**	-	-	-
3.5.1.5. Determine capability deployment locations	1	*							**K**	-	-	-
3.5.1.6. Employ appropriate survey methodologies	2	*							**P**	-	-	-
3.5.1.7. Match capabilities with assigned tasks	2	*							**P**	-	-	-
3.5.1.8. Identify Indicators Of Compromise	2	*							**K**	-	-	-
3.5.1.9. Identify appropriate tools, authorities, methodologies, and existing data sources (e.g., MCT, OSINT, etc.)	2	*							**K**	-	-	-
3.5.1.10. Employ confidence language	3	*							**P**	-	-	-
3.5.1.11. Employ capabilities and methodologies on selected data sources	3	*							**P**	-	-	-
3.5.1.12. Interpret data to extrapolate information (e.g., survey, logs, PCAP, signals, in-phase quadrature, asset lists, topographies, patch history, pipes, etc.)	3	*							**P**	-	-	-
3.5.1.13. Develop visual representations of data	3	*							**P**	-	-	-
3.5.1.14. Explain findings in plain language	4	*							**P**	-	-	-
3.5.1.15. Substantiate findings with evidence	4	*							**P**	-	-	-
3.5.2. ANALYSIS (INTERMEDIATE) **Required Behaviors:** **1. Depicts activities in the operational environment using visualizations to support further analysis or decision-making** **2. Conducts data actions (e.g. clean, transform, and fuse) on existing data sources for enrichment, storage, and integration into information flows** **3. Communicate findings from analysis to support decision-making"** **Criteria:** ***Criteria: Depth of Knowledge: Established practices of all workplace elements***												
3.5.2.1. Fuse multiple visual representations of data	1								-	-	-	-
3.5.2.2. Depict relationships, data flows, and paths using visualization	1								-	-	-	-
3.5.2.3. Develop enriched timelines of activity from two or more data sources	2								-	-	-	-
3.5.2.4. Convert collected data into different formats for analysis tools	2								-	-	-	-
3.5.2.5. Minimize collected data	2								-	-	-	-
3.5.2.6. Develop data pipelines and information flows	2								-	-	-	-
3.5.2.7. Analyze packet captures	2	*							**P**			
3.5.2.8. Validate data sources are informing mission objectives	3	*							**K**	-	-	-
3.5.2.9. Produce analytic reports	3								-	-	-	-
3.5.2.10. Address data gaps in reporting	3								-	-	-	-

Tasks, Knowledge And Technical References	Behavior Match	Core/Cert ^	Deployment * / SEI + CBRN ~	3. Certification For OJT A Tng Start	B Tng Complete	C Trainee Initials	D Trainer Initials	E Certifier Initials	4. Proficiency Codes Used To A 3-lvl IST	B 5-lvl A1C	C 7-lvl SSgt	D 9-lvl SMSgt
3.5.2.11. Make mission decision point recommendations	3								-	-	-	-
3.5.2.12. Prioritize potential targets	3								-	-	-	-
3.5.2.13. Conduct target analysis and development	3	*							K	-	-	-
3.5.3. ANALYSIS (ADVANCED) **Required Behaviors:** **1. Develops and refines workflows to improve decision making** **2. Applies or modifies sophisticated analytical methods to address mission scenarios** **3. Identifies, obtains, and integrates new data sources to refine or improve decision-making** **4. Optimizes or integrates existing data architectures (e.g. sensors, links, storage, processing) to improve mission outcomes** **Criteria:** ***Depth of Knowledge: New practices of all workplace elements***												
3.5.3.1. Synthesize actionable information from large amounts of data	1								-	-	-	-
3.5.3.2. Develop automated tools for analysis	1								-	-	-	-
3.5.3.3. Tailor analytic methods with mission requirements	2								-	-	-	-
3.5.3.4. Leverage complementary analytic approaches within workflows	2								-	-	-	-
3.5.3.5. Validate analytic methods	2								-	-	-	-
3.5.3.6. Recognize bias in workflows	2								-	-	-	-
3.5.3.7. Identify trends and metrics	3								-	-	-	-
3.5.3.8. Extract signatures	3								-	-	-	-
3.5.3.9. Develop data pipelines	3								-	-	-	-
3.5.3.10. Characterize gaps not met by current capabilities	4								-	-	-	-
3.5.3.11. Optimize capabilities to address gaps	4								-	-	-	-
3.5.4. ANALYSIS (EXPERT) **Required Behaviors:** **1. Evaluates, creates, integrates emerging or novel concepts to improve mission efficiency, accuracy, and adaptability** **2. Formulates and establishes organizational analytical techniques to improve mission outcomes** **3. Envisions, advocates for, or engineers data architecture (sensors, links, storage, processing) to optimize or achieve overmatch** **Criteria:** ***New practices/concepts and theories of all workplace elements; is a credible resource in this area***												
3.5.4.1. Evaluate emerging capabilities	1								-	-	-	-
3.5.4.2. Develop new capabilities addressing gaps	1								-	-	-	-
3.5.4.3. Integrate new capabilities into workflows	1								-	-	-	-
3.5.4.4. Institutionalize analytic techniques	2								-	-	-	-
3.5.4.5. Develop curriculum for analytic techniques	2								-	-	-	-
3.5.4.6. Address shortfalls in operations and reporting	2								-	-	-	-
3.5.4.7. Develop exercise scenarios	2								-	-	-	-
3.5.4.8. Cultivate partnerships	3								-	-	-	-
3.5.4.9. Architect solutions to meet requirements	3								-	-	-	-

Section B – Course Objective List

4. There is currently no advanced course. This area is reserved.

Section C - Support Materials

5. The most current products (JQSs/QTPs) can be found at the 81 TRSS/TSQ web page, and are available for download from the web site at https://usaf.dps.mil/teams/10445. Procedures for requesting product development are found in AFI 17-204.

There are currently no support materials. This area is reserved.

Section D - Training Course Index

6. Purpose. This section of the CFETP identifies training courses available for continuation/ supplemental training. For information on all formal courses, refer to the Air Force Education and Training Course Announcements (ETCA) database, at https://usaf.dps.mil/teams/app10-etca/SitePages/home.aspx?isdlg=1

7. Air Force In-Residence Courses.

Course Number	Course Title	Location
WCYBER200	Cyberspace 200	Wright Patterson AFB, OH
WCYBER300	Cyberspace 300	Wright Patterson AFB, OH

8. Operations Training Requirements.

8.1 Operations Training Requirements are identified by the ACC (ACC A3/2/6K) Functional Area Managers for Offensive and Defensive Cyber Operations, IAW ACCI 11-252, para. 1.2.2.

8.2 Joint course requirement and training pipeline information for USCYBERCOM Cyber Mission Force (CMF) Work Roles are located at:
https://intelshare.intelink.gov/sites/uscybercom/Training/SitePages/JCL.aspx

9. Exportable Courses.

For a current list of available online courses on FEDVTE refer to https://fedvte.usalearning.gov/.

Section E - MAJCOM Unique Requirements

10. There are currently no MAJCOM unique requirements. This area is reserved.